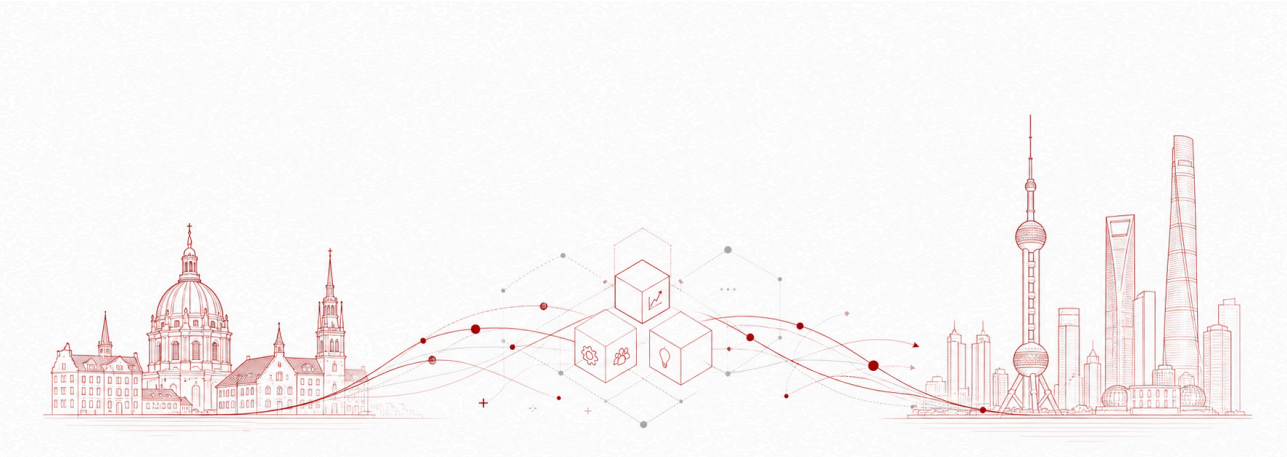
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Danish Companies' Changing Perspectives on China: Implications for Business Model Innovation



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Abstract

Purpose

This study examines how Danish life science companies' perspectives on China evolved between 2021–2025 and how these firms adapt their business models to increasing geopolitical uncertainty and organisational decoupling.

Design

A mixed-methods approach combines longitudinal analysis of the DBOC dataset with qualitative interviews from industry and institutional actors.

Findings

Findings show a shift from optimistic growth expectations to risk-aware engagement, leading to localised activity systems, diversified partnerships, and increased subsidiary autonomy. However, adaptation is constrained by headquarters–subsidiary decoupling and talent challenges.

Implications

Business model adaptation is best understood as continuous activity-system reconfiguration supported by dynamic capabilities rather than one-time strategic change.

Keywords

Business model innovation; Danish companies; China; life science; geopolitical uncertainty; organisational decoupling; resilience

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1. Introduction

1.1 Purpose of the study

The purpose of this study is to analyse how Danish companies' perspectives on China have evolved over the period 2021–2025, using the five-year Danish Business Outlook on China (DBOC) dataset as its empirical foundation. The study aims to identify key shifts in business sentiment over time and to understand how these changes reflect broader transformations in the political, economic, and organisational environment in which Danish firms operate.

Beyond describing developments in business confidence, the study seeks to generate insights into how companies may adapt their business models in response to these evolving conditions. In particular, the research focuses on how prolonged uncertainty, external shocks, and shifting market conditions influence strategic decision-making, organisational coordination, and long-term engagement with the Chinese market. By linking changes in business outlook to questions of adaptation and resilience, the study moves from analysis of perception to consideration of strategic response.

Overall, the purpose of the study is two parts: first, to provide a structured analysis of the evolution of Danish companies' views on China across a critical five-year period and second, to explore how these evolving perspectives can inform business model adaptation in an increasingly complex and uncertain global context.

1.2 Scope of the study

The scope of this study is limited to Danish companies with activities or interests in China, as captured in the Danish Business Outlook on China (DBOC) surveys conducted between 2021 and 2025. The analysis is based on aggregated survey data and focuses on trends in business perceptions across time, rather than on individual firm performance or financial outcomes.

The study examines developments across key dimensions represented in the dataset, including overall business confidence, differences between short-term and long-term outlooks, and variations across company size, sector, and geographic location of respondents (China-based versus Denmark-based). These dimensions are used to identify patterns in how Danish firms interpret and respond to changing conditions in China.

While the dataset covers multiple industries, the analytical scope of the study extends beyond descriptive trend analysis by developing insights into business model adaptation. These insights are derived from the observed evolution in business perspectives rather than from direct firm-level case studies. The study does not attempt to predict future developments in China or to provide a

comprehensive assessment of geopolitical risk. Instead, it focuses on understanding how Danish companies perceive change and how these perceptions may shape strategic adaptation over time.

The temporal scope of the study is strictly confined to the years 2021–2025, a period characterised by significant disruption and adjustment. This defined timeframe allows for a focused analysis of shifts in outlook during a particularly turbulent phase in global business relations.

1.3 Setting the scene

The global business environment has undergone a marked shift in recent years, moving away from deepening globalisation toward a more fragmented landscape in which supply chain reconfiguration, geopolitical competition, and strategic autonomy have become central concerns(European Commission, 2021) . Multinational firms across industries have been compelled to reconsider their international footprints and reassess how they create and capture value across borders(United Nations Conference on Trade and Development, 2023) .

China remains central to this evolving context. Despite growing complexity, it continues to rank among the largest global markets for trade, investment, and innovation(World Bank,2024). Yet the period from 2021 to 2025 introduced new dimensions of uncertainty, testing the assumptions that had long underpinned foreign companies' China strategies. At the same time, China's own economic transition from high-speed growth to domestically driven, high-quality development has reshaped the opportunities and constraints facing foreign firms.

For Danish companies active in China, these global and China-specific dynamics have intersected in particularly consequential ways. The present study draws its empirical basis from the Danish Business Outlook on China (DBOC) Index, which surveys companies affiliated with the Danish Chamber of Commerce in China (DCCC) and the Danish-Chinese Business Forum (DCBF). This group represents a significant segment of Danish commercial engagement with China, operating across multiple sectors with substantial operational stakes. It is within this broader context that the study examines how these companies' perspectives on China have evolved and what these developments imply for strategic adaptation and business model innovation.

1.4 Problem statement

The Danish Trade Council organises its China engagement around four sector teams, reflecting the industries deemed most strategically important to the bilateral relationship: life science and life science, food and agriculture, energy and environment; and maritime and industry. Among these, the life science sector holds a particularly significant position. In 2022, Danish life science exports accounted for 19.7 per cent of total national merchandise exports, making it one of the country's largest export industries (Ministry of Economic Affairs, 2024). Also, between 2008 and 2022, Danish

life science exports to China grew substantially, making China the second-largest destination market for the sector (Ministry of Economic Affairs, 2024). Companies such as Novo Nordisk, Lundbeck, and Novonesis hold well-established positions in the Chinese market, reflecting decades of commercial engagement. This scale of economic weight and China market integration makes life science a natural focal point for understanding how Danish companies navigate and adapt to changing conditions in China.

For consistency, this report uses 'Danish life science companies' as the umbrella term for the empirical focus. The DBOC reports label the relevant category as 'Health/Life Science'. Outside those source-specific labels, the report refers to the sector as life science to align the problem formulation, cases, analysis, discussion, and conclusion.

The decision to narrow this study to the life science sector is further supported by characteristics that make it an instructive case for business model analysis. Life science firms operate at the intersection of commercial activity, public policy, and regulation, placing them at the forefront of the geopolitical and institutional developments that defined the 2021–2025 period. Demand for life science and life science products is also relatively stable and long-term in nature, which limits firms' ability to withdraw quickly from markets such as China and intensifies the pressure to adapt existing operations rather than exit. This combination of strategic economic importance, deep bilateral market integration, and sector-specific adaptation challenges creates a strong analytical fit with the study's focus on business model resilience and organisational decoupling. Despite the strategic importance of Danish life science companies in China, existing research provides limited guidance on how these firms adapt their business models under conditions of prolonged geopolitical uncertainty and organisational decoupling

This study will be guided by the following research question:

How have Danish companies' perspectives on China evolved between 2021-2025, and how can Danish companies in the life science sector adapt their business models to build resilience against geopolitical uncertainty and organisational decoupling?

Sub-Questions:

1. What overall trends and turning points can be identified in Danish companies' perspectives on China in the DBOC dataset from 2021 to 2025?
2. What kinds of risks and challenges from these developments are most relevant for Danish life science companies with Danish headquarters and China-based operations?
3. How can such a company adapt its business model to balance resilience, local responsiveness, and continued China market engagement?

2. Literature Review

This chapter reviews the literature used to understand how Danish life science companies in China may adapt their business models under conditions of macroeconomic, regulatory, and geopolitical uncertainty between 2021 and 2025. Rather than analysing the DBOC survey data directly, the chapter establishes the theoretical and contextual foundation for the later analysis. The review combines established business model literature with more recent developments in business model innovation (BMI) research and studies of uncertainty, adaptation, and organizational change.

2.1 Business Models and Business Model Innovation

A business model describes how a firm creates, delivers, and captures value (Teece, 2010; Zott, Amit, & Massa, 2011). Beyond products and services, business models include customers, value propositions, resources, partners, activities, revenues, and costs. For multinational companies, the business model also includes the geographical organization of activities and the relationship between headquarters and subsidiaries.

Business model innovation (BMI) refers to changes in how value is created, delivered, or captured (Foss & Saebi, 2017). Such changes may involve localization of production, new partnerships, alternative distribution channels, digital solutions, or organizational restructuring. BMI is particularly relevant when firms face environmental disruption because external shocks often affect multiple parts of the business model simultaneously (Teece, 2010).

Recent BMI research has moved beyond viewing innovation as a single strategic change and instead emphasizes continuous adaptation and reconfiguration. Huang, Luo, Xiao, and Dong (2025) distinguish between pioneering BMI and perfecting BMI. Pioneering BMI focuses on creating new markets, technologies, and value propositions, while perfecting BMI focuses on refining existing products, processes, and stakeholder relationships. Their findings suggest that firms increasingly require both forms of BMI simultaneously to balance competitiveness and legitimacy in uncertain environments (Huang et al., 2025). This perspective is particularly relevant for Danish life science companies in China, which must adapt to changing regulations and geopolitical pressures while maintaining market legitimacy and operational efficiency.

2.1.1 Business Models as Storytelling

Magretta (2002) argues that a business model can be understood as a story explaining who the customers are, what they value, how the company creates value, and how it generates profits. She distinguishes the business model from strategy, arguing that the business model explains how the business works, whereas strategy explains how it competes.

This perspective is useful for understanding Danish life science companies operating in China. Their business models are built around international supply chains, local partnerships, regulatory compliance, and cross-border knowledge transfer. Geopolitical tensions, localization requirements, and regulatory changes can disrupt these connections, creating inconsistencies in the business model "story" and forcing firms to rethink how value is created and delivered.

2.1.2 The Four-Element Business Model Framework

Johnson, Christensen, and Kagermann (2008) conceptualize business models through four interdependent elements: Customer Value Proposition, Profit Formula, Key Resources, and Key Processes. Because these elements are interconnected, changes in one area often require adjustments elsewhere.

This framework is particularly relevant in the Chinese life science market. Regulatory requirements, data governance rules, and localization pressures may affect customer value propositions, cost structures, resources, and operational processes simultaneously. Consequently, firms may need to redesign substantial parts of their China business model rather than making isolated adjustments.

2.1.3 Business Model Reconfiguration and Organizational Barriers

Massa and Tucci (2013) distinguish between business model design and business model reconfiguration. While new firms often design business models from scratch, established firms typically modify existing models. However, such reconfiguration is constrained by asset-based barriers, such as existing investments and structures, and cognitive barriers, including managerial assumptions and routines.

For Danish life science companies with long-standing operations in China, these barriers may slow adaptation. Existing investments, organizational routines, and headquarters-driven decision-making can make it difficult to respond quickly to geopolitical uncertainty or regulatory change.

2.1.4 The Business Model as an Activity System

Zott et al. (2011) define business models as systems of interdependent activities that extend beyond firm boundaries and include suppliers, partners, and customers. Business model innovation can therefore involve adding activities, changing activity linkages, or reallocating activities between actors.

This perspective is highly relevant for analysing potential responses to de-risking and geopolitical fragmentation. Firms may localize production, establish parallel supply chains, increase local R&D, or separate Chinese operations from global systems. Such changes represent reconfigurations of the firm's activity system rather than isolated operational adjustments.

2.1.5 Dynamic Capabilities and Continuous Adaptation

Business model theory increasingly emphasizes adaptation as an ongoing capability rather than a one-time strategic decision. Teece (2007) argues that firms require dynamic capabilities to sense environmental changes, seize opportunities, and transform organizational structures accordingly.

This approach is especially relevant for Danish life science companies in China. The period 2021–2025 was characterized by overlapping disruptions including COVID-19 restrictions, supply-chain challenges, geopolitical tensions, and regulatory changes. In such an environment, firms require the capability to continuously adapt their business models rather than merely reacting to individual events.

2.1.6 Uncertainty, Foresight, and Strategic Response

Cuhls (2020) distinguishes between horizon scanning and foresight. Horizon scanning focuses on identifying weak signals of change, while foresight involves interpreting these signals and translating them into strategic action. This distinction is important because many developments in China emerged gradually through policy announcements, regulatory adjustments, and geopolitical shifts.

Similarly, Courtney, Kirkland, and Viguerie (1997) identify four levels of uncertainty ranging from relatively predictable futures to situations characterized by true ambiguity. The Chinese business environment during 2021–2025 often reflected Levels 3 and 4 uncertainty, where firms faced multiple possible futures rather than a single predictable outcome. This suggests that flexibility, resilience, and optionality become critical elements of business model adaptation.

2.1.7 Recent Developments in Business Model Theory

Recent literature increasingly highlights the interaction between business models and external institutional environments. Huang et al. (2025) argue that successful BMI depends not only on internal innovation but also on alignment with strategic orientation and environmental conditions. Their configurational perspective suggests that business model adaptation is most effective when innovation is combined with digital capabilities, regulatory awareness, and sensitivity to changing institutional conditions.

This development extends earlier business model theory by emphasizing that firms do not innovate in isolation. Instead, business model adaptation emerges through the interaction between organizational capabilities, market conditions, regulatory frameworks, and stakeholder expectations. For multinational life science companies operating in China, this implies that successful adaptation requires balancing innovation, compliance, local responsiveness, and global integration.

2.2 Institutional Legitimacy and Headquarters-Subsidiary Coordination

Institutional theory examines how organisations are shaped by the regulatory, social, and normative pressures in their environment (DiMaggio & Powell, 1983; Scott, 2001). For multinational corporations, these pressures operate differently across national contexts. Kostova and Zaheer (1999) argue that headquarters and subsidiaries face competing legitimacy demands: headquarters must protect global standards and strategic coherence, while subsidiaries must conform to local regulations, customer expectations, and government priorities in order to maintain local legitimacy. When the institutional distance between home and host countries is substantial, these divergent pressures can create a structural gap between strategic intent at headquarters and operational reality at the subsidiary level. Kostova and Roth (2002) further show that practice transfer and adaptation within multinational corporations are shaped by both institutional effects from the host environment and relational effects from the headquarters-subsidiary dynamic.

The tension between global integration and local responsiveness has long been recognised in the multinational literature. Prahalad and Doz (1987) argue that multinational firms are subject to competing pressures for global integration, which drives standardisation, and local responsiveness, which demands adaptation to national conditions. Kostova, Marano and Tallman (2016) provide a comprehensive review of fifty years of research on headquarters-subsidiary relationships, tracing how scholarly attention has shifted from structural questions of control and coordination toward more dynamic concerns about responsiveness, entrepreneurship, and strategic adaptation within subsidiaries. More recently, Tse, Meyer, Pan and Chi (2024) reviewed over three hundred studies on multinational strategies in China and identified organisational legitimacy as one of the three persistent institutional challenges facing foreign firms. They note that MNEs in China must develop localised strategies that enhance legitimacy without sacrificing their distinctive competitive advantages, particularly in contexts where isomorphic strategies may conflict with home-country regulations or global standards.

Within this literature, the question of how headquarters-subsidiary dynamics interact with business model adaptation has received growing attention. Mozheiko and Sund (2024) examine how the integration-responsiveness tension manifests as a dual business model trade-off in a Danish multinational's Chinese subsidiary, showing that this trade-off is an ongoing negotiation that intensifies as external conditions become more uncertain. Together, these perspectives provide a theoretical foundation for understanding why headquarters and subsidiaries may experience divergent pressures and how this divergence affects organisational coordination. However, there remains limited research on how these dynamics interact with business model adaptation over an extended period of sustained external uncertainty. This study addresses that gap by examining how the interplay between institutional legitimacy and headquarters-subsidiary coordination shapes business model reconfiguration among Danish life science companies in China.

2.3 Supply Chain Resilience and Geopolitical Risk

The literature on supply chain resilience provides an important complement to business model and institutional theory in understanding how Danish life science companies navigate uncertainty in China. Christopher and Peck (2004) define supply chain resilience as the ability of a supply chain to return to a normal or improved state after a disruption, emphasising that resilience requires both proactive preparation and reactive recovery capabilities. Their framework identifies three sources of vulnerability: internal disruptions, supply-side disruptions, and demand-side disruptions, all of which are relevant to the China context where Danish lifescience companies face operational, regulatory, and market uncertainties simultaneously.

Ivanov and Dolgui (2020) extend this perspective by arguing that supply chain resilience must be understood as "survivability" in intertwined supply networks, where disruptions in one node can cascade through the entire system. This is particularly relevant for Danish life science companies whose operations in China are embedded in global value chains spanning R&D, manufacturing, procurement, and distribution. The COVID-19 pandemic and subsequent geopolitical tensions exposed the fragility of such intertwined networks, making resilience not merely an operational concern but a strategic imperative that shapes business model design.

Caldara and Iacoviello (2022) provide a quantitative framework for measuring geopolitical risk, defining it as the potential occurrence, realisation, and escalation of adverse events associated with warfare, terrorism, and political tensions among countries. Their geopolitical risk index captures the type of uncertainty that Danish life science companies in China have faced since 2021, including US-China trade tensions, the war in Ukraine, and EU-China regulatory disputes. For this study, geopolitical risk is not treated as an external variable but as a structural condition that permeates business model decisions, from sourcing strategies to R&D localisation to subsidiary autonomy.

The combination of supply chain resilience and geopolitical risk literature helps explain why business model adaptation in the China context is not simply about optimising efficiency but about ensuring operational continuity under conditions of systemic uncertainty. As Christopher and Peck (2004) note, resilient supply chains require redundancy, flexibility, and collaboration, qualities that must be embedded in the business model itself rather than treated as peripheral operational adjustments.

2.4 Research Gap

While existing literature addresses business model innovation, resilience, and institutional pressures separately, limited research examines how these factors interact under sustained geopolitical uncertainty. In particular, there is a lack of understanding of how external pressures such as geopolitical risk and organisational decoupling translate into changes in specific business model

components. This study addresses this gap by linking external uncertainty to internal adaptation mechanisms and business model reconfiguration.

For understanding business models, resilience, regulation, geopolitical risk, and multinational coordination. However, there is still a need for more research on how business model thinking changes over time when companies face several external shocks at once. Foss and Saebi (2017) argue that business model innovation research needs more knowledge about what triggers change and under what conditions it happens.

The main theoretical argument of this study is that business model innovation in China should be understood as an adaptive response to a changing environment. The business model is not fixed; it is shaped by market opportunities, supply-chain risk, regulation, geopolitical uncertainty, policy priorities, and the relationship between headquarters and local operations. This adaptive perspective draws on Teece's (2007) dynamic capabilities framework, which emphasises that firms must continuously sense environmental shifts, seize emerging opportunities, and transform their resource configurations to sustain competitive advantage. In the context of Danish life science companies in China, this means that business model resilience is not achieved through a single strategic pivot but through the cumulative development of organisational capabilities that enable ongoing reconfiguration of value creation, delivery, and capture mechanisms.

This gap matters specifically for the research question because existing BMI research gives limited guidance on how resilience is built when geopolitical uncertainty and organisational decoupling occur simultaneously. In the China context, external shocks do not only pressure market strategy; they also reshape headquarters-subsidiary coordination, local responsiveness, and the allocation of activities across borders. By connecting DBOC sentiment trends with an in-depth Novonesis case, supported by Novo Nordisk and institutional interviews, this study examines how business model adaptation becomes a response to the three elements named in the problem formulation: resilience, geopolitical uncertainty, and organisational decoupling.

By combining industry-level sentiment data with illustrative organisational insights, this study examines how business model adaptation unfolds across Danish life science companies in China under conditions of sustained uncertainty.

2.5 Literature Analysis with Scopus

Scopus was selected because it provides broad coverage of peer-reviewed academic literature across management, international business, innovation, supply-chain, and social science fields. Searches were conducted using Scopus advanced search syntax in the article title, abstract, and keyword fields through the TITLE-ABS-KEY function. The search strategy combined broad conceptual terms with narrower contextual terms related to China, multinational enterprises, subsidiaries, life science, life sciences, Danish companies, de-risking, and geopolitical uncertainty. This layered strategy was chosen because the exact topic of Danish life science companies in China is highly specific, while the theoretical foundations come from broader academic debates.

The search process began with broad thematic searches on business model adaptation under uncertainty, dynamic capabilities, institutional legitimacy, supply-chain resilience, and horizon scanning. These searches were then narrowed using contextual terms such as China, multinational enterprises, subsidiaries, life science, and de-risking. In addition, seed-article searches were used to verify whether key academic sources already cited in the literature review and theoretical framework were indexed in Scopus. This helped ensure that the literature base was not only relevant but also traceable.

Sources were included when they contributed directly to the project's research question or sub-questions, offered a foundational concept used in the theoretical framework, or helped explain the empirical context of multinational firms operating in China under uncertainty. Sources were excluded when they were not available in English, were outside the project's conceptual focus, addressed China only in a generic way, or did not contribute to the project's analytical framework. Foundational sources published before the main contemporary period were retained when they defined concepts that remain central to the study, such as business models, dynamic capabilities, institutional legitimacy, and headquarters-subsidiary coordination.

The Scopus search was also used as a quality-control step. It made the literature collection process more transparent by documenting search strings, filters, screening logic, and selection criteria. Non-academic sources, including DBOC reports, government publications, World Bank reports, Reuters, MERICS, and policy documents, were not treated as part of the Scopus-based academic literature. Instead, they were used as empirical or contextual sources. This distinction supports research integrity by making clear which sources are peer-reviewed academic literature and which sources are used to describe market, policy, regulatory, or historical developments.

The full search strings, selection criteria, and search log are provided in Appendix A.

3. Theoretical Framework

3.1 Business Model Canvas

As discussed vastly in the literature review, the concept of the business model has become central to understanding how firms create, deliver, and capture value. Osterwalder and Pigneur (2010) define a business model as the logic of how an organization operates and generates value through interconnected activities and resources. Their Business Model Canvas (BMC) breaks this logic into nine key components, including value propositions, customer segments, channels, customer relationships, revenue streams, key resources, key activities, key partnerships, and cost structures.

In this study, the Business Model Canvas is not used as a prescriptive or applied managerial instrument. Instead, it serves as a conceptual structure that allows for the systematic identification and categorization of core business model components. By decomposing the business model into distinct elements, the BMC enables an analytical examination of how different parts of a firm's value creation architecture respond to external pressures.

Importantly, the modular nature of the BMC supports the analysis of differentiated vulnerability across business model components. For example, key partnerships and supply chains may be particularly exposed to geopolitical fragmentation, while value propositions or customer relationships may require adaptation in response to shifting demand conditions. This analytical use of the BMC allows for a structured investigation into which components are critical for resilience.

3.2 Horizon Scanning

Horizon scanning can be conceptualized as a forward-looking analytical framework that enables organizations to systematically identify and interpret emerging changes in their external environment. It is commonly defined as a structured process through which weak signals, trends, and potential disruptions are detected at an early stage, allowing decision-makers to anticipate developments before they become fully materialized (Hines et al., 2019). Unlike traditional forecasting approaches, which often rely on extrapolating historical data, horizon scanning explicitly focuses on uncertainty, discontinuity, and novelty, thereby offering a more flexible approach to understanding complex and evolving environments.

Horizon scanning is not only used as a singular method, but rather a composite framework consisting of multiple analytical practices. These may include environmental scanning, trend analysis, and expert consultation, all of which contribute to the identification and interpretation of early-stage developments (Cuhls, 2015). The value of this framework lies not only in detecting change, but in structuring how such change is perceived and integrated into strategic thinking. In doing so, horizon scanning provides a systematic way to engage with uncertainty rather than attempting to eliminate it.

In academic literature, horizon scanning is frequently associated with the ability to detect “weak signals,” which are early indicators of potentially significant change (Hines et al., 2019). Although such signals may initially appear marginal or uncertain, they can evolve into critical drivers of transformation over time. The challenge, therefore, is not simply to observe the environment, but to recognize which developments may gain significance in the future.

This forward-looking capability is particularly important in environments where change unfolds gradually. As highlighted by Wintle et al. (2020), horizon scanning allows organizations to explore emerging issues that are not yet fully recognized, thereby enhancing preparedness for uncertain developments. In this sense, horizon scanning contributes to a shift from reactive to proactive strategic behavior. Rather than responding to disruptions once they occur, organizations can begin adapting at earlier stages, when strategic flexibility is still high.

To structure the temporal dimension of change identified through horizon scanning, the Three Horizons framework provides a useful conceptual extension. This framework distinguishes between three overlapping time horizons, each representing a different stage of development and a different level of uncertainty.

The first horizon refers to the present dominant system, encompassing existing structures, practices, and models that currently define how value is created and maintained. This horizon is characterized by relative stability and a focus on efficiency, optimization, and incremental improvement. Because the structures within this horizon are well established, uncertainty is comparatively limited. However, this stability can also mask underlying vulnerabilities, particularly in systems that are highly optimized and therefore less adaptable to external shocks. Horizon scanning within this scope focuses primarily on identifying short-term risks and gradual changes that may affect current operations.

The second horizon represents a transitional phase in which emerging developments begin to challenge the existing system. In this horizon, elements of the current system coexist with new and evolving practices, creating tension between established structures and potential alternatives. This phase is inherently more uncertain, as it involves processes of transformation that are not yet fully defined. According to Cuhls (2015), horizon scanning plays a particularly important role in this phase by identifying trends and shifts that signal the beginning of structural change. These may include evolving market dynamics, technological developments, or geopolitical shifts that gradually reshape the environment in which organizations operate.

The third horizon captures long-term developments and possible future systems that have not yet become dominant but may fundamentally alter existing structures. This horizon is characterized by high uncertainty, as it involves early-stage innovations, weak signals, and potential discontinuities. As noted by Wintle et al. (2020), horizon scanning in this context focuses on identifying emerging issues and exploring their possible implications, even when their future impact remains unclear. Although

developments within this horizon may initially appear distant, they are crucial for understanding long-term transformation and for shaping strategic positioning over time.

3.3 The Business Model as an Activity System

Zott and Amit (2010) and Zott, Amit and Massa (2011) conceptualise the business model as a dynamic system of interdependent activities and identify three mechanisms through which such systems change: adding new activities, linking activities in new ways, and changing which actor performs a given activity. The present section focuses on the analytical properties of this perspective that make it a useful framework for examining organisational adaptation in contexts characterised by institutional complexity and cross-border coordination challenges.

The activity system perspective offers several analytical advantages for studying how multinational corporations restructure their operations when facing external disruptions. First, its relational focus makes it possible to examine not just what activities a company performs but how those activities are connected to activities performed by other actors in its broader ecosystem, including headquarters, local subsidiaries, external partners, and suppliers. When cross-border coordination becomes difficult, the connections between these actors may weaken, shift, or require alternative arrangements. The activity system lens makes these relational changes visible and analysable. Second, the framework accommodates both incremental and radical change, allowing analysis of minor adjustments to individual activities as well as fundamental restructuring of the entire activity system. This is relevant in environments where external pressures vary in intensity and where companies may adapt gradually rather than through sudden transformation. Third, the perspective is inherently dynamic, suited to studying how business models evolve over an extended period rather than examining them as static snapshots at a single point in time.

A distinctive feature of the activity system perspective is its attention to the connections between activities rather than to the activities themselves. In the context of multinational corporations, the connections between headquarters and foreign subsidiaries are central to how value is created and delivered. When these connections are disrupted by regulatory divergence, geopolitical tensions, or changing market conditions, the activity system must be reconfigured in order to maintain operational coherence. The three mechanisms identified by Zott, Amit and Massa (2011) provide a structured vocabulary for describing such reconfiguration: adding locally embedded activities that were previously performed at a distance, reconfiguring the linkages between headquarters and subsidiary activities, and reallocating the performance of specific functions from one organisational actor to another. These properties make the activity system perspective particularly suited to examining how multinational firms manage the tension between maintaining globally integrated operations and developing locally responsive capabilities.

3.4 Institutional Legitimacy and the Dual Business Model Trade-off

The headquarters-subsidiary relationship in multinational corporations is shaped by institutional pressures that operate differently at the global and local levels. Kostova and Zaheer (1999) argue that multinational firms face a fundamental legitimacy challenge: they must simultaneously conform to the institutional expectations of their home country environment and those of each host country in which they operate. Institutional legitimacy refers to the generalised perception that the actions of an entity are desirable, proper, or appropriate within a given social and institutional context. When the institutional environments of the home and host countries differ substantially, the pressures placed on headquarters and on the local subsidiary can diverge significantly. Headquarters must protect global standards, strategic coherence, and brand integrity, while the subsidiary must respond to local regulatory demands, customer expectations, and government priorities in order to maintain local legitimacy. When these competing pressures become irreconcilable, a gap emerges between strategic intent at headquarters and operational reality on the ground. This gap is commonly conceptualised in the literature as organisational decoupling.

Mozheiko and Sund (2024) provide an empirical examination of this theoretical insight through their study of a Danish multinational corporation's Chinese subsidiary. They demonstrate how multinational firms manage a dual business model trade-off: the tension between maintaining a globally integrated business model and developing a locally responsive one. Their analysis reveals that this trade-off is not a static strategic choice but an ongoing negotiation that intensifies when external conditions become more uncertain. In the Chinese context, regulatory changes, geopolitical pressures, and evolving market expectations continually reshape the conditions under which the subsidiary operates, forcing repeated adjustments to the balance between global integration and local adaptation. Their findings illustrate how the dual business model trade-off manifests in concrete organisational decisions regarding which activities are performed locally, which are controlled centrally, and how the two systems are coordinated.

Taken together, these two perspectives provide a complementary framework for understanding the headquarters-subsidiary relationship under institutional complexity. Kostova and Zaheer (1999) explain why decoupling occurs: the divergence between the institutional pressures facing headquarters and those facing the subsidiary creates a structural tension that cannot be fully resolved through better communication or management practice alone. Mozheiko and Sund (2024) explain how decoupling manifests: as a gradual reconfiguration of the relationship between the global and local business models, in which certain activities are increasingly performed locally while others retain their global character. In combination, these frameworks offer a theoretically grounded vocabulary for analysing both the institutional roots and the operational consequences of headquarters-subsidiary dynamics in multinational corporations.

3.5 The Use of Theories in This Project

In this project, the theories are not used separately but are combined into one overall framework. The purpose of this is to create a clearer structure for understanding business models and how different theoretical perspectives relate to each other.

At the centre of this framework is the Business Model Canvas (BMC). The BMC is used as a structuring model that divides the business model into different components, such as value propositions, customer segments, key activities, and partnerships. In this project, the BMC works as a link between the general idea of a business model and the more detailed theories included in the framework. It provides a simple and systematic way of identifying what elements make up a company's business model.

The BMC also makes it possible to look at business models as something that can be broken down into smaller parts. This allows for a more structured understanding of how different elements are related, without treating the business model as one single concept. In this way, the BMC acts as a middle layer that connects broader theory with more specific perspectives.

To support the BMC, additional theories are included to give more depth to certain parts of the model. In particular, the framework puts emphasis on key activities, as these are central elements within the business model structure.

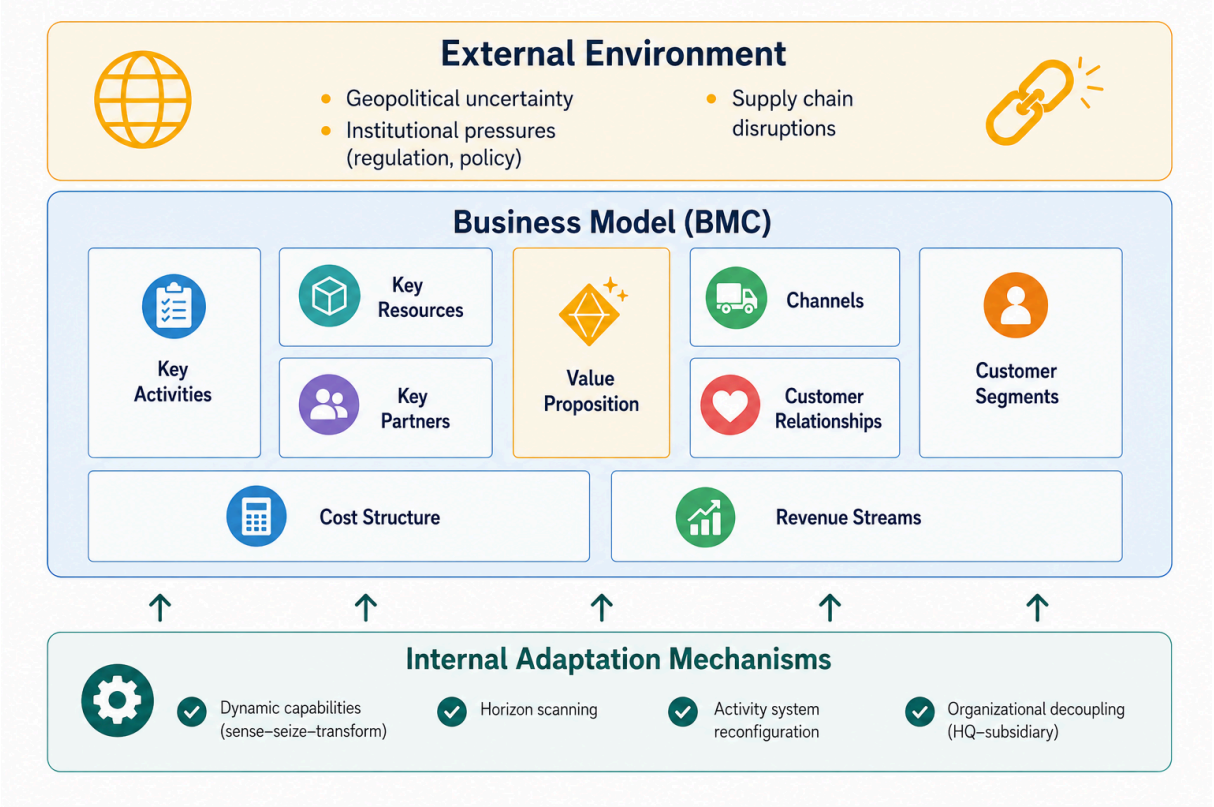
The activity system perspective (Zott & Amit, 2010; Zott et al., 2011) focuses on how activities are connected to each other and form a system. It provides a way of understanding how different activities within a business model are linked together rather than existing independently.

The dynamic capabilities framework (Teece, 2007) introduces a time-related perspective, focusing on how firms and their capabilities can be understood over time. It adds a processoriented dimension to the overall framework.

In addition, institutional theory and headquarters–subsidiary perspectives are included to reflect the organizational and external context in which firms operate. These perspectives focus on the role of institutional environments and organizational structures, particularly in multinational settings.

Overall, the BMC works as the main structuring framework, while the other theories provide additional perspectives. More specifically, dynamic capabilities explain how firms adapt key resources and key activities over time, while the activity system perspective explains how these activities are reconfigured through adding, linking, and reallocating. Institutional theory and headquarters–subsidiary perspectives explain constraints on customer relationships and organisational coordination, while horizon scanning supports the adaptation of value propositions and strategic positioning under uncertainty. that help describe and organise specific parts of the business model. The revised framework below makes this hierarchy more explicit by linking the BMC components most

relevant to the study, key activities, key partnerships, key resources, channels/customer relationships, and cost/revenue logic, to activity-system reconfiguration, dynamic capabilities, and the external institutional and geopolitical pressures shaping adaptation.



Model 1. *The model shows that external pressures create the need for adaptation, internal organisational mechanisms determine how firms respond, and these responses translate into reconfiguration of specific business model components (Source: AI Generated).*

4. Contextual Setting

This chapter presents the quantitative foundation for the study by introducing the Danish

Business Outlook on China (DBOC) dataset and explaining why the life science sector (DBOC: Health/Life Science) is used as the empirical focus. The purpose is not to provide a full historical account of every external event affecting China between 2021 and 2025, but to establish the data context needed for the later analysis of business model adaptation, organisational decoupling, and strategic resilience. A fuller overview of the contextual developments is provided in Appendix C

4.1 What Happened in the World in the Past Five Years?

The period from 2021 to 2025 was marked by overlapping global and China-specific disruptions that reshaped the business environment for foreign companies. These developments form the contextual background for interpreting Danish companies' changing outlook on China and for understanding why business model adaptation became increasingly relevant. A more detailed event-by-event overview is provided in Appendix C.

In 2021 and 2022, the most immediate disruption came from COVID-19 restrictions, especially China's prolonged lockdowns, quarantine rules, and travel limitations. These restrictions affected market access, business travel, customer contact, supplier coordination, and communication between headquarters and China-based subsidiaries. In 2022, the World Bank highlighted how China's economy was affected by COVID-19 outbreaks and lockdowns, particularly during the March-May period (World Bank, 2022a). For multinational companies, the inability to move people across borders made local teams more important and increased reliance on remote coordination.

At the same time, global supply chains became more fragile. Port delays, transport shortages, production disruptions, and changing demand patterns made international sourcing and delivery less predictable. The World Bank linked lockdowns in China and global supply-chain disruption to wider economic uncertainty (World Bank, 2022c). This exposed the risks of highly efficiency-oriented supply chains and encouraged companies to reconsider supplier dependency, regional production, inventory levels, and backup options.

The Russian invasion of Ukraine in February 2022 added another layer of uncertainty. Although not directly connected to China, the war affected energy prices, inflation, logistics, food prices, and overall global business confidence. The World Bank warned that the war in Ukraine, combined with China-related lockdowns and supply-chain disruption, increased risks of global slowdown and stagflation (World Bank, 2022c). For companies, this meant higher costs and greater uncertainty around pricing, production, and investment planning. It also strengthened the broader shift from globalisation as efficiency toward resilience as a strategic priority.

China's reopening at the end of 2022 created expectations of recovery in 2023. Business travel resumed, customer contact became easier, and many companies expected Chinese demand to rebound. However, the recovery proved weaker and more uneven than expected. The World Bank later pointed to weak domestic demand, property-sector stress, and softer consumption as key constraints on China's recovery (World Bank, 2024; World Bank, 2025). This challenged earlier assumptions that China would automatically return to high-growth market conditions.

During the same period, regulation became more important. China's Data Security Law and Personal Information Protection Law came into effect in 2021, increasing attention to data governance, cybersecurity, personal information protection, and cross-border data transfer (National People's

Congress, 2021a; National People's Congress, 2021b). The revised Counter-Espionage Law further increased attention to national security and information risks (Library of Congress, 2023). This is especially relevant for life science companies, because they often work with clinical information, patient-related data, digital health solutions, and regulatory documentation. Regulation therefore became not only a compliance issue, but also a business model issue affecting how information, R&D, and digital activities can be organised.

Geopolitical tensions also intensified. In Europe, the policy language increasingly shifted from full "decoupling" to "de-risking", meaning that companies and governments did not necessarily seek to leave China, but wanted to reduce strategic dependencies and manage exposure more carefully (MERICS, 2023). At the same time, Chinese official sources continued to present China as open to foreign investment and emphasised measures to expand market access, support fair competition, and facilitate innovation-related factors for foreign investors (State Council Information Office, 2024). This created pressure for companies to diversify suppliers, rethink China-only production or sourcing, and develop more flexible regional strategies while still maintaining access to the Chinese market.

Finally, China's longer-term policy direction remained important. The upcoming 15th Five-Year Plan and continuing life science reform signal Chinese priorities around technological self-reliance, advanced manufacturing, green transition, life science innovation, and high-quality growth. Chinese official life science sources show continued policy attention to medical insurance coverage, affordable medicines, and support for innovative drugs during the 14th Five-Year Plan period (State Council Information Office, 2025). For Danish life science companies, this creates both opportunity and pressure: firms may benefit if their value propositions align with Chinese policy priorities, but doing so may require local partnerships, local R&D, local production, and China-specific adaptation.

Overall, the 2021-2025 period should be understood as a transition from growth-oriented optimism toward a more cautious, risk-aware, and strategically selective engagement with China. The key issue is not only that external shocks occurred, but that these shocks forced companies to reconsider how they organise value creation, delivery, and control across borders.

5. Methodology

5.1 Philosophical Assumptions

This project is grounded in an interpretivist philosophy. Interpretivism holds that social reality is not a fixed objective structure but is constructed through human experience and perception. This philosophical starting point is appropriate because the central phenomenon we are studying, how Danish companies perceive and respond to changing conditions in China, is inherently subjective and context dependent. The DBOC data supports this choice directly. Throughout the dataset, respondents based in China and respondents based in Denmark consistently report different outlooks on the same market, demonstrating that business reality in this context is experienced differently depending on position, proximity, and organizational role.

5.2 Research Approach

The project follows an abductive research approach. Abduction occupies a middle ground between deduction, which tests predefined hypotheses, and induction, which builds theory purely from empirical observation(Dubois & Gadde, 2002) . In abductive reasoning the researcher moves iteratively between empirical patterns and theoretical frameworks, using each to refine the other(Kovács & Spens, 2005) . This approach fits the project because we begin with observable patterns in the DBOC data, identify theoretical frameworks that can explain those patterns, and then refine our understanding through primary interview data. The gap between Denmark-based and China-based respondents throughout the dataset, for example, prompted us to draw on headquarters-subsidiary theory, which is then tested and contextualized through interviews(Gioia et al.,2013).

5.3 Research Strategy

The project combines a mixed methods strategy with an inspired multiple case study design. The reason for the inspired and not fully integrated multiple case design is that the project does not look narrowing and focused down one or two cases. This study looks at both of the empirical industry data as important knowledge and insight, but does not go deep enough into detail and research about the cases to call it a traditional multiple case study. Therefore this research strategy is inspired by. The mixed methods approach integrates quantitative data from the DBOC index with qualitative data from semi-structured interviews, treating them as complementary rather than parallel. The DBOC data provides longitudinal breadth across five years and approximately 50 to 80 Danish companies per quarter, while the qualitative data provides organizational depth through the lived experiences of specific managers and industry representatives. The design focuses is a in depth analysis on Danish-headquartered life science companies with localized operations in China. This is appropriate

because Sub-Q2 and Sub-Q3 require understanding of organizational strategy and decision making within a specific institutional context that aggregate survey data alone cannot provide.

The project adopts a mixed-methods industry-level research design combining longitudinal survey data with qualitative interview insights. The Danish Business Outlook on China (DBOC) dataset serves as the primary empirical foundation, capturing industry-level developments across Danish companies operating in China over time.

The qualitative component does not follow a traditional case study design. Instead, interviews with Novonesis and Novo Nordisk are used as illustrative empirical windows into the Danish life science industry, providing organisational depth and helping to interpret patterns observed in the DBOC data. The purpose is not to analyse or compare individual companies, but to contextualise broader industry-level dynamics of business model adaptation.

5.4 Research Quality

5.4.1 Reliability

The DBOC index's consistent question format and institutional backing across all 18 quarters supports replication and reduces measurement variation over time. However, quarterly sample sizes ranging from approximately 40 to 86 companies introduce some variation in representativeness between periods, which should be considered when interpreting quarter-to-quarter shifts.

Interview reliability is supported by the use of a standardized guide for case company interviews and consistent gioia coding procedures. The variation in interview format (open-ended for institutional interviews, structured for case companies) reflects their distinct purposes but may introduce comparability constraints.

5.4.2 Validity

The DBOC index applies a consistent standardized methodology across all 18 quarters from Q1 2021 to Q4 2025, which makes longitudinal comparisons reliable. The survey is conducted jointly by the Embassy of Denmark, DCCC, DCBF, and SDC, lending institutional credibility to the data. However, the sample is restricted to DCCC and DCBF member companies and the index measures directional sentiment rather than absolute performance, meaning findings should be interpreted as indicative of perceived trends rather than precise quantitative outcomes.

Interview validity is strengthened by triangulation across data sources (DBOC trends, institutional perspectives, and company-level accounts) and methods (longitudinal quantitative sentiment analysis and qualitative thematic analysis). Theoretical frameworks, specifically Teece (2007) on dynamic capabilities and headquarters-subsidiary theory informed the coding structure and interpretation of organizational decision-making patterns.

5.4.3 Triangulation

Triangulation operates across both sources and methods. DBOC longitudinal trends provide aggregate sentiment direction; institutional interviews (DCBF, DCCC) contextualize macro-level dynamics; and case company interviews (Novonesis, Novo Nordisk) examine operational responses. Discrepancies between Denmark-based and China-based respondent outlooks in the DBOC data are examined against interview accounts of headquarters-subsidiary communication gaps.

5.5 Ethical considerations

Participants were informed of the purpose of the project before agreeing to participate. Consent to record interviews was obtained explicitly. Participants were given the option to remain anonymous in the final project if preferred. Organizational roles and strategic details are preserved in transcripts where they are essential to analytical validity, but personal identifiers are removed. Participants will review transcripts prior to final publication. Strategic details that could identify specific individuals within organizations are retained only where they contribute directly to the research argument and with participant awareness.

5.6 Participants

To support both industry-level context and firm-level insight, this research draws on four qualitative interviews with representatives from key Danish business organisations and companies operating in or engaging with China. These participants were selected to provide a balanced perspective, combining macro-level institutional knowledge with micro-level corporate experience.

The Danish-Chinese Business Forum (DCBF)

The Danish-Chinese Business Forum (DCBF) is a network-based organisation that aims to facilitate cooperation between Danish and Chinese stakeholders with shared commercial interests. Its primary purpose is to promote the exchange of information, knowledge, and professional connections among companies, organisations, and executives engaged in Danish-Chinese business activities (Danish-Chinese Business Forum, n.d.)

Through its platform, DCBF brings together private firms, public actors, and industry experts, thereby supporting knowledge sharing, relationship-building, and market understanding across the two countries. The forum plays an important role in reducing informational barriers and strengthening cross-cultural business collaboration by providing a space for dialogue on market conditions, regulatory developments, and strategic opportunities (Danish-Chinese Business Forum, n.d.)

In this research, the interview with DCBF is used to provide industry-level background, offering insights into the overall landscape in which Danish companies operate in China.

The Danish Chamber of Commerce in China (DCCC)

The Danish Chamber of Commerce in China (DCCC) is an official, non-profit organisation established to promote trade, investment, and commercial relations between Denmark and China. Founded in 2000 and approved by Chinese authorities, DCCC is one of the recognised foreign chambers of commerce operating in China (Danish Chamber of Commerce in China, n.d.)

Within this study, the interview with DCCC serves as a complementary industry-level perspective, contributing structured knowledge on the regulatory environment, market dynamics, and strategic considerations relevant to Danish firms.

Novo Nordisk

Novo Nordisk is a Danish multinational healthcare company founded in 1923 and headquartered in Denmark. The company focuses on the prevention and treatment of serious chronic diseases, including diabetes, obesity, and other metabolic disorders (Novo Nordisk, n.d.).

With operations in more than 170 countries and a workforce exceeding 69,000 employees, Novo Nordisk combines scientific research, innovation, and long-term healthcare strategies to improve global health outcomes (Novo Nordisk, n.d.).

China represents a strategically important market for the company, where it has established significant manufacturing and supply capabilities and continues to invest in expanding its presence to meet growing healthcare demand (China Daily, 2026).

In this research, the interview with Novo Nordisk provides firm-level insight from a large multinational corporation, particularly in relation to how global healthcare companies navigate the Chinese market and adapt their strategies to local conditions.

Novonesis

Novonesis is a global biosolutions company formed through the merger of Novozymes and Chr. Hansen, combining expertise in biotechnology and industrial bioscience. The company develops biological solutions, including enzymes, microbes, and probiotics, which are applied across industries such as agriculture, food production, energy, and health (Novonesis, n.d.).

The organisation’s mission is to harness the power of biology to address global challenges by improving efficiency, sustainability, and performance across value chains. Its solutions are designed to support areas such as food security, environmental sustainability, and industrial optimisation (Novonesis, n.d.).

Novonesis positions itself as a key actor in the emerging bioeconomy, emphasising innovation and scientific advancement as drivers of long-term economic and environmental value (Novonesis, n.d.).

In this study, the interview with Novonesis contributes firm-level insight from the biosolutions and industrial biotechnology sector, offering a complementary perspective to Novo Nordisk and highlighting differences in industry dynamics and strategic approaches.

5.7 Data Collection

The project draws on two categories of data. Secondary data consists of the 18 quarterly DBOC reports published between Q1 2021 and Q4 2025, providing a standardized longitudinal record of Danish business sentiment across four market dimensions, six sectors, and multiple company size and location categories. Primary data is collected through semi-structured interviews with participants from DCCC, the Danish Embassy, and the case company. Semi-structured interviews are chosen because the research questions require understanding of reasoning, perception, and strategic thinking rather than simple factual responses. The format provides sufficient structure to address consistent themes across respondents while allowing flexibility to explore unexpected but analytically relevant answers. Interviews will be conducted in English, recorded with participant consent, and subsequently transcribed for thematic analysis.

A consistent semi-structured interview guide was used for both case companies to ensure comparability; the institutional interviews (DCBF, DCCC) followed a more open-ended format to capture macro-level perspectives and inform case selection. Interviews were conducted primarily in English, with occasional Danish remarks retained where they occurred in the natural conversational flow. Interview durations ranged from approximately 38 to 85 minutes. All participants provided explicit consent for recording, transcription, and anonymization. Transcripts will be shared with participants for review prior to publication.

5.8 Data Analysis of Interviews

The coding of the interviews to identify themes will be carried out using a Gioia-inspired method. Although the Gioia method (Gioia, Corley & Hamilton 2013 15) is rooted in grounded theory studies, it can, with some adaptation, support a critical realist analysis. While Gioia originally emphasizes concept development and the participants' voice, the method is here used to systematize the interviews and form the basis for identifying underlying mechanisms.

The approach to coding in the Gioia methodology is based on initially focusing on the informants' own statements without interpreting them. At this stage, we are dealing with our raw empirical transcribed interview data. Subsequently, underlying themes are identified through coding across the different interviews, and the informants' statements are grouped into categories that we assess as similar. In practice, this categorization was conducted after an initial rough coding, after which the

coded quotes were divided into different colors representing corresponding thematic connections. After creating clusters of differently colored quotes with formulations within the same area, a further summary was carried out in what, in Gioia terminology, is referred to as second-order dimensions. Here, summaries and formulations are developed based on the expressed statements. Finally, these formulations are synthesized into overarching themes, which in Gioia are called aggregated dimensions. It is on the basis of these themes, which also function as analyzed organizational challenges, that the analysis is structured.

This method was chosen because it supports the inductive approach of the project while also providing a clear overview of the coding, ensuring a high level of transparency in both the data and the coding process for use in the analysis.

5.9 Data Analysis of DBOC Data

The analysis operates on three levels corresponding to the three sub-questions. First, a longitudinal trend analysis of the DBOC data identifies the major phases, turning points, and patterns across 2021 to 2025, directly addressing Sub-Q1. DBOC data was analyzed through descriptive trend analysis, with particular attention to divergences between Denmark-based and China-based respondent groups; Health/Life Science sector responses were examined against the full sector sample to contextualize sector-specific dynamics. Second, Gioia analysis of interview transcripts identifies how managers and institutional representatives understand geopolitical risk, decoupling, and business model adaptation, addressing Sub-Q2 and Sub-Q3. Third, triangulation across both data sources examines where interview findings confirm, contradict, or add nuance to what the DBOC data suggests at the aggregate level.

5.10 Limitations of the Dataset

The DBOC sample is restricted to DCCC and DCBF member companies, meaning findings may not fully represent all Danish companies active in China. As a diffusion index it captures sentiment direction rather than absolute business performance.

The interview sample is small and purposive given the limited company access; findings are therefore analytically generative rather than statistically generalizable. The Novo Nordisk interview is further limited by the participant's recent arrival in China (one year of tenure), reducing historical depth for the 2021–2025 period. The group's position as SDC students in Beijing facilitated access but may introduce response bias, particularly in institutional interviews where participants expressed interest in leveraging the research for advocacy purposes. This study was however deliberately scoped around a small, purposive interview sample combined with longitudinal survey data, prioritising analytical depth within a specific institutional context over statistical generalisability in multiple industries.

6. Cases and Data Overview

This chapter presents the empirical setting for the analysis. It first describes the DBOC dataset as a secondary, quarterly business-sentiment index covering Danish companies in China from 2021 to 2025, including sector, quarter, company size, and respondent-location dimensions. It then introduces the qualitative interview material and clarifies the role of the interviews. Novonesis as the more detailed interview insight and Novo Nordisk as additional supporting perspective supporting triangulation.

The company interviews are not treated as standalone cases, but as empirical illustrations that provide organisational insight into broader patterns identified in the DBOC data.

6.1 DBOC Health/Life Science Sector Overview

6.1.1 2021: A Lagging but Resilient Sector

At the launch of the index in Q1 2021, Health/Life Science was identified, together with Agriculture/Food, as one of the largest export categories for Danish companies in China, yet both sectors reported lower-than-average improvements in outlook compared to the overall index of 55.9. This suggests that even during the most optimistic period of the dataset, the sector's largest and most established exporters were comparatively cautious, while the wider rebound in sentiment was driven more by smaller, less export-intensive categories. The same pattern persisted into Q2, when the overall index held steady at 55.6; the modest improvement recorded that quarter was again attributed primarily to the Agriculture/Food and Design/Lifestyle sectors rather than to Health. As the aggregate index turned negative in Q3, falling to 48.7, Health/Life Science followed the general downward trend, though it was Energy and Environment & Water that stood out as the only sectors remaining in positive territory. By Q4 2021, as the index recovered marginally to 49.9, Health was one of the few sectors reported as holding an unchanged outlook rather than a deteriorating one, and was grouped among the four of six sectors showing an upward trend compared to the previous quarter. Taken together, the 2021 data positions Health/Life Science as a sector that neither led the early optimism nor suffered the sharpest reversals, instead tracking the aggregate index with a persistent lag.

6.1.2 2022: Below the Index Throughout the Downturn

The severe contraction documented for 2022 as a whole is mirrored, and in some respects amplified, within the Health/Life Science sector. While Q1 2022 saw five of six sectors deteriorate without Health being separately singled out, from Q2 onward the reports explicitly note that the sector indicated levels below the overall index, a pattern that held as the aggregate index collapsed to its dataset low of 39.4. This sub-index positioning continued through Q3, when Health and Design were the only two sectors reported below the overall index of 41.8, even as Energy, Agriculture/Food,

Logistics, and Environment & Water all posted scores above it. The trend extended into Q4 2022, when Health again indicated levels below the overall index of 39.9, confirming that the sector experienced the COVID lockdowns, supply chain disruption, and the Russian invasion of Ukraine described in the yearly overview not merely in line with the aggregate downturn, but somewhat more acutely than the average Danish company operating in China at the time.

6.1.3 2023: A Volatile Recovery

The fragile and uneven recovery characterising 2023 at the aggregate level is visible in an even more pronounced form within Health/Life Science. In Q1 2023, as the overall index rebounded sharply to 54.8, all sectors improved, including Health, though the report does not single it out as an exceptional performer relative to Environment & Water, which reached a historic high. This improvement proved short-lived: in Q2, Health re-entered deteriorating index scores after the prior quarter's gains, even as the aggregate index remained in positive territory at 50.6. The reversal deepened in Q3 2023, when the aggregate index fell into negative territory for the first time that year, to 47.1; the report explicitly attributes this decline to deteriorating scores in Logistics, Design/Lifestyle, and especially Health, marking the sector as a principal driver of the broader downturn rather than a passive follower of it. By Q4 2023, Health had reversed course again, recording one of the sector-level increases that quarter alongside Logistics and Design, even though Energy remained the only sector in positive territory overall. This sequence of improvement, reversal, and partial recovery within a single year illustrates a volatility in the Health sector that exceeds the swings of the aggregate index, foreshadowing the divergence between short-term sentiment and longer-term sector positioning that becomes more pronounced from 2024 onward.

6.1.4 2024: Persistent Instability Amid Aggregate Stabilisation

Although the aggregate DBOC Index stabilised close to the neutral 50-point level throughout 2024, the Health/Life Science sector continued to exhibit the quarter-to-quarter instability already evident in 2023. Q1 2024 saw Health, together with Design, indicate a deteriorated outlook, even as Environment, Energy, Logistics, and Food all improved. The sector reversed again in Q2 2024, this time appearing among the three sectors with improved outlooks alongside Logistics and Energy. The improvement was again temporary: by Q3 2024, Health/Life Science had entered negative territory once more after this single improved quarter, while the sectors reporting positive short-term outlooks that quarter, Agriculture & Food, Environment & Water, Logistics & Maritime, and Energy, did not include Health. The Q4 2024 report does not single out Health/Life Science by name among either the five sectors with negative outlooks or the sole sector (Energy) in positive territory, suggesting the sector's position that quarter was not distinctive enough to warrant separate comment, consistent with a broader pattern of the sector oscillating around, rather than decisively above or below, the aggregate trend.

6.1.5 2025: A Tentative Stabilisation

The pronounced swings in aggregate sentiment described for 2025, from the early-year high of 52.9 in Q1, through the sharp Q2 decline to 49.0, to the steadier recovery across Q3 and Q4, find a somewhat calmer counterpart within the Health/Life Science sector. In Q1 2025, five of six sectors reported improved short-term outlooks, with only Design/Lifestyle deteriorating, implying that Health/Life Science was among the sectors carried upward by the broader rebound in confidence that quarter. This improvement did not survive the mid-year downturn: in Q2 2025, Design/Lifestyle was the only sector to buck the prevailing decline, meaning Health/Life Science declined in step with most of the dataset, consistent with the widespread concerns over decoupling and headquarters-subsidiary communication highlighted in the yearly overview. By Q3 2025, as the aggregate index began its recovery to 50.78, Health/Life Science was described as merely trailing the previous quarter's sentiment, that is, broadly flat rather than sharply moving in either direction, while Energy, Logistics/Maritime, and Agriculture/Food posted the more pronounced positive upticks. The year's final quarter brought a renewed improvement: in the Q4 2025 report (published January 2026), Health/Life Science was named among the sectors with an increasing outlook, together with Agriculture/Food, Design/Lifestyle, and Environment/Water, even as Energy and Logistics/Maritime declined. This closing improvement positions the sector, by the end of the five-year period, as cautiously aligned with the aggregate stabilisation described in the yearly summary, rather than as either a leading indicator or a clear laggard.

6.2 Interview Data and Case Profiles

This section presents the interview material from four semi-structured interviews: two company interviews and two institutional interviews. The purpose of this descriptive overview is not to make statistically representative claims about all Danish companies in China, but to show how actors with direct knowledge of Danish life science activities interpret the changes shown in the DBOC data. The interviews therefore add organisational depth to the DBOC material: while the DBOC data shows when sentiment changed, the interviews help explain how these changes are experienced and translated into business model decisions. The full Gioia coding structure and selected supporting interview evidence are provided in Appendix B.

Novonesis is the longest and most in depth interview in this study. The company provides the richest basis for analysing business model adaptation because the interview offered longer historical coverage of China operations, direct discussion of headquarters-subsidiary relations, and detailed reflections on localisation, production, innovation, and organisational coordination across the 2021-2025 period.

The interview with Novo Nordisk provides a second life science perspective on regulatory pressure, market access, price-volume trade-offs, local competition, and China-based operations, but the shorter interview.

The DCBF interview provides a broader institutional perspective on the geopolitical and competitive pressures facing Danish life science companies in China. It is mainly used to contextualise the company interviews, especially around US pricing pressure, Chinese industrial policy, and growing local competition.

The DCCC interview provides a China-based institutional perspective on the DBOC data and Danish business sentiment. It helps explain differences between Denmark-based and China-based respondents, especially regarding headquarters' slower reaction time, local autonomy, and the importance of understanding market developments from within China.

The interview material was analysed using a Gioia-based coding logic. First-order concepts were kept close to interviewee language, including terms such as "closed loop model", "China speed", "more autonomy", "communication loops", "volume versus price", and "seeing is understanding". These were grouped into second-order themes such as localised business model adaptation, headquarters-subsidiary friction, risk-aware commitment, life science-specific regulatory pressure, and talent-based knowledge transfer. The full Gioia coding structure is included in Appendix B, Table B.1, while selected supporting quotes are included in Appendix B, Table B.2. In the main analysis, the findings are summarised into five aggregate patterns.

The combined Gioia analysis identified six main aggregate themes across the interviews. Overall, the interview material indicates that China remains strategically important for Danish life science companies, but the market is now experienced as more complex, competitive, and uncertain.

The strongest theme was risk-aware commitment and strategic optionality. Interviewees did not describe leaving China as the main response to uncertainty. Instead, they framed de-risking as diversification, long-term planning, and keeping strategic options open while maintaining a presence in China.

A second key theme was China as a strategically necessary but more contested market. China is still important because of its market size, innovation capacity, R&D environment, and life science demand. However, local competition, pricing pressure, state support, and policy shifts make the market more difficult for foreign companies.

The third theme was life science and life-science specific adaptation pressures. The interviews showed that companies are affected by sector-specific issues such as patent expiry, product lifecycles, market access, procurement, clinical development, patient access, IP, data regulation, and price-volume trade-offs.

The fourth theme was localised business model reconfiguration. Companies are adapting through local R&D, production, partnerships, digital channels, acquisitions, and regional innovation models.

However, these changes are mostly practical and incremental rather than complete business model transformations.

The fifth theme was headquarters-subsidiary decoupling and decision friction. China-based teams often understand local changes faster than headquarters, creating friction around autonomy, communication, risk perception, and decision-making. Local responsiveness therefore depends on stronger information flows and clearer decision authority.

The final theme was talent, tacit knowledge, and cultural translation. The findings highlight the importance of local talent, experienced leadership, and employees who can translate between Chinese market realities and Danish/global headquarters expectations.

Overall, the interviews suggest that Danish life science companies' perspectives on China have become more differentiated. China is still viewed as strategically important, but the sources of value and risk have changed. The interview material refines the DBOC analysis by showing that the shift from optimism to cautious stabilisation is produced through selective localisation, stronger attention to local partnerships, negotiation between headquarters control and subsidiary autonomy, and de-risking through diversification rather than exit. For the research question, the key implication is that business model adaptation in China is not a single strategic move, but an ongoing organisational capability: companies must remain locally responsive while preserving global coordination and reducing exposure to geopolitical and organisational decoupling. Additional supporting quotes and coding evidence for these findings are provided in Appendix B.

7. Analysis

The company interviews are used here as illustrative empirical material and are not analysed as individual cases; instead, they support the analysis of broader industry-level mechanisms.

This chapter presents the empirical analysis of how Danish life science companies have navigated evolving conditions in China between 2021 and 2025, and how these experiences inform business model adaptation. Drawing on the DBOC sentiment data presented in Chapter 3 and the qualitative interviews conducted with industry representatives and corporate managers, the analysis examines the mechanisms through which external disruptions are translated into internal organisational responses. The analysis is structured around the theoretical frameworks introduced in Chapter 2, integrating the Business Model Canvas as an analytical decomposition tool with the activity system perspective, institutional legitimacy theory, and horizon scanning.

The analysis proceeds in five sections. The first section examines organisational decoupling and the headquarters-subsidiary tension. The second section analyses activity system reconfiguration through the mechanisms of adding, linking, and reallocating activities. The third section applies the Three Horizons framework to strategic positioning. The fourth section addresses supply chain resilience and risk diversification. The fifth section examines the emerging importance of talent and cultural alignment. The chapter concludes with a synthetic discussion of how these elements cohere into an adaptive business model pattern for the sector.

7.1 Organisational Decoupling and the Headquarters-Subsidiary Tension

The DBOC dataset reveals a persistent and widening divergence in sentiment between Denmark-based and China-based respondents throughout the 2021–2025 period. While this gap narrowed marginally after 2022, it remained structurally significant, with headquarters respondents consistently slower to internalise both negative shocks and recovery signals. This pattern corresponds closely to the phenomenon of organisational decoupling described by Kostova and Zaheer (1999), wherein headquarters and subsidiaries face competing institutional legitimacy demands that create a structural gap between strategic intent and operational reality.

The interview data confirms that this decoupling is not merely a perceptual lag but an active coordination challenge. As one institutional representative observed, "the sentiment in Denmark is slower to react to the reality on the ground in China… if you see something in one quarter in China, then it takes about one to two quarters for it to reflect in the data in Denmark" (Participant 3). This delay is compounded by governance structures in which "HQ needs to go through their formal committees and they have to form an opinion about it as a part of a broader global context… China just feeds in as one of them" (Participant 3). The decoupling is therefore institutionalised through

decision-making routines that filter local signals through global strategic frameworks, diluting their urgency.

For the life science sector, this decoupling carries specific risks. life science companies operate at the intersection of commercial activity and stringent regulatory compliance, where delayed decision-making can affect product registrations, clinical trial protocols, and government reimbursement negotiations. The DCBF representative emphasised that headquarters are "becoming more and more concerned. But the question of what they are actually doing about it is a different matter," noting that even when urgency is created in boardrooms, "the next day, everyone is completely loaded with their regular workload and nothing changes" (Participant 4). This suggests that decoupling in Danish life science firms is characterised not by a lack of information, but by a failure to translate information into organisational action.

Mozheiko and Sund (2024) conceptualise this as the dual business model trade-off: the ongoing negotiation between maintaining globally integrated operations and developing locally responsive capabilities. The interview data reveals that this trade-off manifests differently across firms. Novo Nordisk's China affiliate operates what the manager described as a "closed loop model," with R&D, production, and commercial activities all localised, suggesting a high degree of operational integration (Participant 1). However, even this structure retains significant HQ control: "every affiliate has some kind of freedom, and at the same time… headquarters have some kind of control… it's kind of always this kind of trading of the power… changing over time, changing with the management style" (Participant 1). This "trading of power" illustrates Mozheiko and Sund's (2024) argument that the dual business model trade-off is not a static strategic choice but an ongoing negotiation.

Novonesis presents a different configuration. The company restructured its regional marketing to report directly to local sales leadership rather than global marketing, creating "no additional layer of decision making between them" (Participant 2). Yet this empowerment remains bounded: "if the sales team… request a price that is lower than the global standard, they still have to apply it to the global marketing business unit for them to decide whether it's okay or not" (Participant 2). The resulting "power bargaining between region and the global headquarter" (Participant 2) confirms that decoupling is not resolved by structural redesign alone, but requires continuous negotiation.

From a Business Model Canvas perspective, decoupling primarily affects Key Activities and Customer Relationships. When HQ-subsidiary coordination lags, the ability to adapt customer-facing activities such as pricing negotiations, regulatory submissions, and partnership development is constrained. The DCCC representative noted that "companies who truly give autonomy to their Chinese division… they tend to have better results" (Participant 3), suggesting that reducing decoupling requires not just better communication but genuine delegation of Key Activities. This reflects institutional theory and

headquarters–subsidiary dynamics, highlighting how organisational decoupling shapes business model components such as customer relationships and key activities.

7.2 Activity System Reconfiguration: Adding, Linking, and Reallocating

Zott and Amit (2010) and Zott, Amit and Massa (2011) identify three mechanisms through which business model activity systems change: adding new activities, linking activities in new ways, and changing which actor performs a given activity. The interview data reveals that Danish life science companies in China have engaged in all three mechanisms, though with varying degrees of intentionality and success.

7.2.1 Adding Local Activities

The most visible form of activity system reconfiguration is the addition of locally embedded activities that were previously performed at a distance. Both Novo Nordisk and Novonesis have expanded their China operations beyond sales and distribution to include R&D and manufacturing. Novo Nordisk's manager emphasised that the company has "R&D also in China, we have production also in China, and these affiliate or commercial parts also in China," describing this as a "closed loop model" (Participant 1). Similarly, Novonesis has "two enzyme production sites in China" and is "considering licensing out part of our production to external partners" (Participant 2).

This expansion of Key Activities and Key Resources represents a shift from an export-oriented business model toward a locally integrated one. As the DCBF representative noted, the distinction between "exporter to China" and "produce and sell locally in China" is critical because "the risks and pressures are quite different" (Participant 4). For exporters, currency depreciation and trade friction pose direct threats to revenue streams; for local producers, the challenge shifts to cost competition and regulatory compliance.

7.2.2 Reconfiguring Linkages

The second mechanism, linking activities in new ways, is evident in how companies are restructuring their innovation processes. Novonesis has developed a "regional innovation" model specifically for China, which focuses not on inventing new enzyme backbones but on "taking enzyme products off the shelf and developing new applications to serve regional customer needs" (Participant 2). This "solution innovation" rather than "enzyme innovation" allows the company to bypass the early-stage R&D bottleneck and move faster to market. The model also enables parallel global and China innovation tracks: for CO2 capture, the company partnered with a Danish technology provider globally while selecting a different Chinese partner locally, establishing "a parallel check" (Participant 2).

This reconfiguration of Key Partnerships and Key Activities aligns with Huang et al.'s (2025) distinction between pioneering BMI and perfecting BMI. The global-Danish partnership represents pioneering exploration, while the local-Chinese partnership represents perfecting adaptation. By

running both tracks simultaneously, Novonesis attempts to balance global strategic coherence with local responsiveness.

7.2.3 Reallocating Activities

The third mechanism: Changing which actor performs an activity, is visible in the localisation of management and the shifting boundaries between HQ and subsidiary. The DCCC representative observed that "a lot of foreign managers are being switched out with Chinese managers… now you have Chinese employees who have worked… been working with the companies for 10, 15 years now and have a really good understanding of the companies" (Participant 3). This reallocation of human resources from expatriate to local talent changes the nature of the activity system by embedding local institutional knowledge more deeply within the firm.

However, this reallocation also introduces new tensions. Novonesis's manager noted increasing difficulty in attracting high-quality technical talent, particularly because the company's labs are concentrated in Beijing while qualified candidates may prefer Shanghai or other cities. This creates a structural recruitment bottleneck that could constrain the firm's ability to scale its local R&D activities (Participant 2). The "people problem", where tacit knowledge resides with an aging workforce and knowledge transfer mechanisms are underdeveloped thus emerges as a critical barrier to activity system reconfiguration. This aligns with the activity system perspective, demonstrating how firms reconfigure key activities and partnerships through adding, linking, and reallocating activities within the business model.

7.3 Horizon Scanning and Strategic Positioning

The Three Horizons framework (Cuhls, 2020; Wintle et al., 2020) provides a useful lens for understanding how Danish life science companies are sequencing their adaptation efforts across different timeframes. The DBOC data shows that sentiment stabilised near neutral levels by 2024–2025, suggesting that firms have moved from reactive crisis management toward more deliberate strategic positioning.

7.3.1 Horizon 1: Protecting the Core

In the immediate term, companies are focused on defending existing revenue streams and operational efficiency. For Novo Nordisk, this means managing the forthcoming patent expiry on semaglutide, which the manager characterised as "business as usual… nothing new… every single patent in every single market has time of expiring" (Participant 1). The strategic response is a classic volume-versus-price calculation: "how much volume versus how much price… price went down, we are increasing volume, that's always the name of the game" (Participant 1). This reflects Horizon 1 thinking, optimising the existing dominant system.

The DCBF representative framed this more urgently, nothing that Chinese competitors are preparing generic versions and that "China will become so cost-competitive, from a pricing perspective that it will come like a bulldozer and squeeze the Danish life science industry" (Participant 4). From this perspective, Horizon 1 efficiency gains are necessary but insufficient; they must be accompanied by structural adaptation.

7.3.2 Horizon 2: Transitioning Activities

The intermediate horizon involves developing new practices that challenge the existing system while still leveraging current capabilities. The most prominent Horizon 2 development is the shift toward B2C and digital channels. The DCCC representative highlighted that "Novo Nordisk has adopted… a plan to do B2C… basically, they had previously been distributors, but now they're building up a big social media platform to be able to sell directly to consumers" (Participant 3). This represents a fundamental change in Channels and Customer Relationships within the Business Model Canvas, moving from a traditional B2B pharmaceutical model toward direct patient engagement.

Similarly, both Novo Nordisk and Novonesis are deepening their local R&D footprint, treating China not merely as a sales market but as an innovation hub. The DCCC representative noted "an increasing trend in terms of viewing China as a R&D destination… trying to set up R&D strands and research centres in China" (Participant 3). For Novonesis, regional innovation allows the company to "skip over the early stage to search for new enzyme backbone… and start from what is available already on the shelf" (Participant 2), dramatically shortening development cycles.

7.3.3 Horizon 3: Long-Term Transformation

The third horizon involves positioning for possible future systems that may fundamentally alter existing structures. Here, the alignment with China's 15th Five-Year Plan emerges as a critical strategic variable. Chinese five-year planning creates a predictable policy trajectory that firms can align with, particularly in life science, sustainability, and advanced manufacturing. The DCBF representative emphasised that life science is a priority area in China's industrial policy, and that companies should "look seriously at opportunities to work with China, so that at least Denmark has a foot in the door and remains part of the future landscape" (Participant 4).

Novo Nordisk's strategy of acquiring Chinese biotech companies "looking… to acquire these Chinese companies… not only to rely on our own R&D" (Participant 1) represents a Horizon 3 positioning move. By integrating Chinese innovation into its global pipeline, the company attempts to capture value from China's rising R&D capabilities rather than merely defending against them. This aligns with Teece's (2007) dynamic capabilities framework: sensing the shift in global innovation geography, seizing opportunities through acquisitions, and transforming the firm's resource configuration.

However, Horizon 3 positioning is constrained by geopolitical risk. The DCBF representative warned that "if a Danish company enters a licensing agreement with a Chinese company, it would make no sense to try to sell the resulting product in the US" due to political sensitivities, and that "clinical trials conducted in China will not be recognised in the US and potentially not even in the EU" (Participant 4). This creates a strategic dilemma that Courtney, Kirkland and Viguerie (1997) would classify as high-level uncertainty: firms face multiple possible futures in which the value of China-centric R&D may be high or low depending on geopolitical trajectories. This reflects the role of horizon scanning in enabling firms to anticipate change and adapt their value propositions and strategic positioning under conditions of uncertainty

7.4 Supply Chain Resilience and Risk Diversification

Christopher and Peck (2004) argue that supply chains designed primarily for efficiency become vulnerable when unexpected disruptions occur. Ivanov and Dolgui (2020) extend this by emphasising that disruptions in one node of an intertwined network can cascade throughout the system. The interview data reveals that Danish life science companies have responded to these insights by diversifying their geographic footprint, though with varying strategic logics.

Novonesis explicitly frames its China presence as a risk diversification strategy: "You never put all your eggs into one basket… you actually put it in several or many" (Participant 2). The company serves more than 30 industries globally, allowing it to "cultivate bets and all drivers for future business" (Participant 2). Within China, it has expanded from one to two production sites and is increasing "solution innovation" investment. Crucially, the company rejects exit as a response to competitive pressure: "Because the local competition is growing, we should stay here rather than escaping from here. We need to be part of this society and really follow what's going on here" (Participant 2). This "stay to learn" strategy treats the China operation as a strategic intelligence asset that can "share this with other regions before those Chinese players go into their region" (Participant 2).

Novo Nordisk's approach is similarly rooted in diversification but framed in terms of global competitiveness rather than China-specific risk: "You need to have this diversification… diversify yourself in many, many ways all over the world in order to get the best from all parts of the world. If you keep all your production in Europe, and all your R&D in Europe, it will be very hard to compete" (Participant 1). This suggests that for large pharmaceutical firms, China integration is not merely about managing China risk but about accessing resources, R&D talent, manufacturing scale, clinical trial speed that are increasingly concentrated in China.

The DBOC data supports this interpretation. The R&D sentiment index consistently outperformed sales sentiment from 2024 onward, suggesting that companies were maintaining long-term strategic commitments even as short-term market conditions deteriorated. This divergence between short-term

sales pessimism and long-term R&D optimism reflects a supply chain resilience strategy that prioritises capability maintenance over immediate revenue optimisation.

From a Business Model Canvas perspective, these adaptations affect Key Partnerships, Key Activities, and Cost Structure. Localising production and R&D changes the partnership architecture. Shifting from Danish suppliers and research institutions to Chinese contract manufacturers, biotech firms, and academic partners. It also alters cost structures, as Novonesis noted: "China is very well known with its highly efficient manufacturing ability. We have to keep our cost leading position in China" (Participant 2). This highlights how external pressures influence the reconfiguration of key partnerships and cost structures within the business model to enhance resilience.

7.5 Talent, Tacit Knowledge, and Cultural Alignment

An emerging theme across the interviews, though not fully theorised in the existing literature review, is the growing importance of human capital and cultural alignment in managing HQ-subsidiary coordination. The group working notes identify a "people problem": an aging workforce with deep tacit knowledge is approaching retirement, raising questions about how knowledge is transferred and how qualified replacements are developed.

The interview data validates this concern. Novonesis's manager explicitly identified talent as "the biggest future challenge": "We are hiring more and more positions in China, and I found it more and more difficult actually to attract very high quality talents to our company" (Participant 2). The constraint is structural: the company's labs are in Beijing, but "there are many, many talents in Shanghai labor market. But does it mean that I have to establish the same lab in Shanghai or in Nanjing or in Guangzhou, Shenzhen, everywhere?" (Participant 2). This geographic concentration of infrastructure creates a talent-market mismatch that could limit the firm's ability to scale its regional innovation model.

The DCCC representative similarly highlighted that the decline in expatriate managers, while reducing costs, has created a "cross-cultural management gap" because "the number of young Danish managers coming out to China with that cultural context of both China and Denmark has decreased significantly" (Participant 3). This decline in bicultural managers may actually exacerbate decoupling, as these individuals historically served as boundary spanners who could translate between HQ and subsidiary institutional logics.

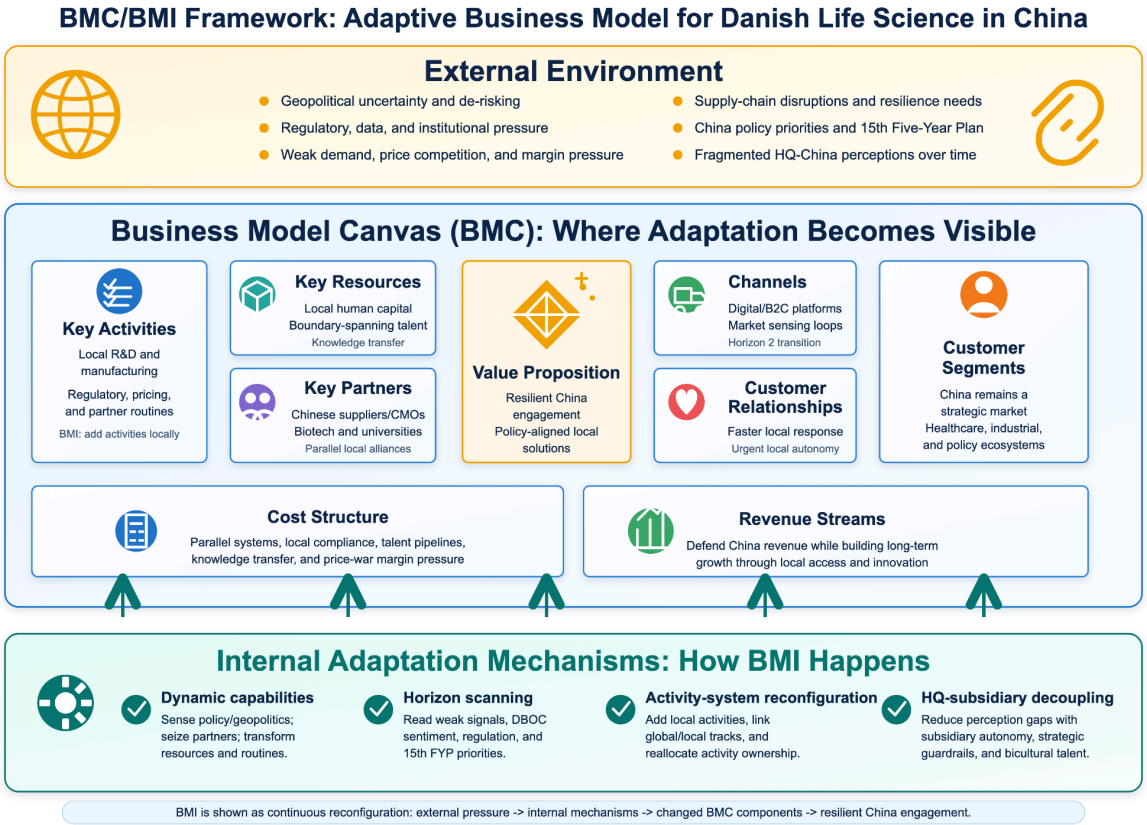
The working notes' observation that "culture development (alignment) in a firm to get an understanding of each side narrow down decoupling" finds empirical support in Novonesis's emphasis on shared culture elements: "explore, ownership, care, impact" that are intended to align behaviour across regions despite linguistic and institutional differences (Participant 2). However, the same manager acknowledged that "not everyone in the headquarters could… adapt better to the regional

practice… they need a lot of communication loops to really agree on something" (Participant 2). This suggests that cultural alignment programmes may reduce but cannot eliminate the structural tensions inherent in HQ-subsidiary coordination. This demonstrates how internal organisational capabilities influence the effectiveness of business model adaptation, particularly in relation to key resources and coordination.

7.6 Synthesis: Toward an Adaptive Business Model for Danish Life Science in China

Taken together, the analysis reveals that Danish life science companies are engaged in a multi-layered adaptation process that spans activity systems, governance structures, and strategic horizons. The DBOC sentiment curve from 2021 optimism through the 2022 crisis to 2024–2025 stabilisation reflects not merely external market fluctuations but an organisational learning process. Firms have moved from reactive crisis management toward more deliberate business model reconfiguration, though with significant variation in speed and depth.

Using the theories as previously disclosed in the theory chapter and shown in the circular model. After integrating and applying background theories to the collected data, it will be shown that the adaptation pattern can be summarised through four interconnected shifts in the Business Model Canvas.



Model 2. *The model applies the BMC/BMI framework to the study's findings, showing how external pressures and internal adaptation mechanisms translate into concrete changes across key business model components, including localised activities, reconfigured partnerships, digital channels, customer relationships, local resources, cost structures, and revenue logic (Source: AI Generated).*

First, Key Activities are being localised, with R&D and manufacturing increasingly performed within China rather than exported from Denmark. This reduces currency and trade risks but exposes firms to local cost competition.

Second, Key Partnerships are being reconfigured, with global-Danish innovation partnerships supplemented by local-Chinese alliances, acquisitions, and contract manufacturing relationships. This reflects a shift from bilateral HQ-subsidiary coordination to multilateral ecosystem engagement.

Third, Customer Relationships and Channels are evolving from traditional B2B models toward direct consumer engagement and digital platforms, enabled by China's advanced e-commerce infrastructure.

Fourth, Key Resources particularly human capital are being reallocated from expatriate to local management, though this transition creates talent attraction and knowledge transfer challenges that may constrain adaptation speed.

These shifts are not occurring in isolation but as interdependent components of a broader activity system reconfiguration (Zott et al., 2011). The localisation of manufacturing, for example, enables the localisation of R&D, which in turn facilitates local partnerships, which strengthen government relationships, which improve market access. Conversely, failure in one domain such as persistent HQ-subsidiary decoupling can constrain progress in others, as delayed decisions prevent the timely allocation of resources to local innovation.

The analysis also reveals an important tension between two adaptation logics. On one hand, firms are pursuing defensive strategies aimed at protecting core revenues and reducing geopolitical exposure, what the DCBF representative termed "defending themselves for as long as possible" (Participant 4). On the other hand, firms are pursuing offensive strategies aimed at capturing opportunities from China's innovation ecosystem, acquiring local biotech firms, licensing Chinese-developed molecules, and aligning with industrial policy priorities. Novo Nordisk's manager captured this duality: "You can call this diversification, you can also call de-risking… using resources all over the world… China has really huge resources and in some way very advanced resources compared to the rest of the world" (Participant 1).

The ability to sustain both logics simultaneously depends on dynamic capabilities (Teece, 2007) specifically, the capacity to sense geopolitical and competitive shifts, seize emerging opportunities through rapid organisational reconfiguration, and transform internal structures to support new activity

patterns. The interview data suggests that Danish life science companies have developed strong sensing capabilities; they are aware of competitive threats and policy shifts. However, their seizing and transforming capabilities remain weaker: headquarters response remains slow, talent pipelines are constrained, and governance structures retain significant friction. As Novonesis's manager summarised, "we need to be empowered even more, and we need to reduce the communication costs between the region and the global organization" (Participant 2).

This finding has direct implications for the research question. Danish companies' perspectives on China have evolved from market optimism through crisis-driven caution toward a more differentiated, risk-aware engagement. Business model adaptation is occurring, but it is uneven constrained by organisational decoupling, talent bottlenecks, and geopolitical ambiguity. The following chapter discusses how these findings relate to the broader literature and what they imply for managerial practice and policy. Taken together, these findings illustrate how business model adaptation emerges through the interaction between external pressures and internal mechanisms, resulting in the reconfiguration of multiple interdependent business model components.

8. Discussion

This study adopts an industry-level mixed-methods approach rather than a traditional case study design. The DBOC dataset provides the secondary empirical foundation, while interviews with the 2 institutional actors, together with Novonesis and Novo Nordisk serve as illustrative organisational and industry primary insights that support and contextualise the broader patterns identified in the data. The purpose is not to analyse individual firms, but to better understand industry-level mechanisms of business model adaptation. This framing ensures analytical consistency between the methodology, analysis, and discussion, positioning company evidence as illustrative rather than as the basis for case generalisation.

8.1 Situating the Findings

Building on the analysis, the discussion moves from empirical findings to theoretical interpretation, focusing on how Danish life science companies' evolving perspectives on China are translated into business model adaptation under conditions of geopolitical uncertainty and organisational decoupling.

The preceding analysis traced how Danish life science companies have adapted their China operations between 2021 and 2025. It showed that adaptation was not a single strategic pivot but a layered process of organisational learning, activity system reconfiguration, and governance recalibration. This chapter situates those findings in relation to the theoretical frameworks introduced in Chapter 2 and the empirical patterns documented in the Danish Business Outlook on China (DBOC) dataset. The aim is to explain not only what changed in Danish life science companies' China business models, but also why those changes took the form they did and what they imply for business model innovation (BMI) theory more broadly.

The discussion follows the case framing introduced in the methodology: Novonesis and Novo Nordisk is used for empirical data and triangulation. The DBOC dataset provides the longitudinal sector-level context, and the institutional interviews with DCCC and DCBF clarify how broader market and policy developments are interpreted by Danish business actors. Therefore, company-level claims are anchored primarily in Novonesis and novonordisk and where institutional interviews point in the same direction.

This study does not adopt a comparative case study like design. Instead, it analyses industry level developments among Danish life science companies in China, using interview insights from 2 industry institutional actors, along Novonesis and Novo Nordisk as illustrative examples to support and contextualise patterns identified in the DBOC dataset.

The discussion is organised around the concentric framework developed for this study. At the centre lies the Business Model Canvas (BMC), which serves as the analytical hub for identifying how value creation, delivery, and capture are changing. Surrounding the BMC is the activity system, understood through the mechanisms of adding, linking, and reallocating activities (Zott & Amit, 2010; Zott, Amit & Massa, 2011). This activity system is animated by dynamic capabilities, the capacity to sense, seize, and transform (Teece, 2007) and shaped by the headquarters-subsidiary relationship. Both are embedded in a broader institutional context of regulatory, geopolitical, and market pressures (Kostova & Zaheer, 1999). Reading the findings through these layers makes it possible to connect micro-level organisational adjustments with macro-level environmental shifts.

8.2 Business Model Adaptation as Activity System Reconfiguration

A first contribution of this study is to show that BMI in the China context is best understood as activity system reconfiguration rather than as a discrete redesign of one or two business model components. The analysis identified four interconnected shifts in the Business Model Canvas: the localisation of Key Activities, the reconfiguration of Key Partnerships, the digitisation of Channels and Customer Relationships, and the reallocation of Key Resources toward local human capital. These shifts did not occur independently, rather, they reinforced one another in a cumulative pattern of local embeddedness.

This finding extends Zott and Amit's (2010) activity system perspective by showing how the three mechanisms of change: adding, linking, and reallocating activities operate under conditions of sustained external uncertainty. Adding local R&D and manufacturing activities enabled firms to reduce currency exposure and trade-friction risk, but it also required new linkages with local suppliers, contract manufacturers, and biotech partners. Reallocating activities from expatriate to local managers changed who performed boundary-spanning roles, which in turn affected how headquarters and subsidiaries were linked. The interdependence of these mechanisms supports the argument that business models are systems in which adjustment in one element creates pressure for adjustment elsewhere (Johnson, Christensen & Kagermann, 2008).

The study also nuances Huang et al.'s (2025) distinction between pioneering and perfecting BMI. Rather than choosing between exploration and exploitation, the interviewed firms pursued both simultaneously. Novo Nordisk's acquisition of Chinese biotech assets and Novonesis's regional innovation partnerships represent pioneering moves into new innovation geographies, while local cost optimization, digital B2C channels, and efficiency improvements in manufacturing represent perfecting activities. This dual pursuit suggests that in high-uncertainty host markets, successful BMI requires ambidexterity not only over time but also across organisational levels and geographic units.

8.3 The Institutional Roots of Organisational Decoupling

The persistent divergence between Denmark-based and China-based respondents in the DBOC data is one of the study's most striking empirical patterns. The analysis interpreted this divergence as organisational decoupling: a growing structural gap between strategic intent at headquarters and operational reality in the subsidiary. This interpretation is consistent with Kostova and Zaheer's (1999) argument that multinational enterprises face competing legitimacy demands across institutional environments. Headquarters must protect global coherence, compliance standards, and brand integrity, while the Chinese subsidiary must respond to local regulation, government expectations, and customer practices.

Mozheiko and Sund (2024) describe this tension as a dual business model trade-off that intensifies under uncertainty. The interview data confirms and extends this insight in two ways. First, the trade-off is not only between global integration and local responsiveness, but also between speed and control. Subsidiaries perceive market changes faster than headquarters, yet formal governance routines delay the translation of local signals into decisions. As one interviewee noted, headquarters must process China-specific developments within a broader global portfolio, which flattens the urgency of local shocks. Second, the trade-off is dynamic rather than structural. The balance of power between headquarters and subsidiary shifts with management changes, product life cycles, and external events, requiring continuous negotiation rather than a one-off organisational design.

The life science sector magnifies the consequences of decoupling because of its regulatory intensity. Delayed decisions in drug registration, clinical trial design, reimbursement negotiation, or pricing can have long-lasting competitive effects. This means that decoupling is not merely an abstract coordination problem, it is a risk to operational continuity and market positioning. The finding therefore supports Prahalad and Doz's (1987) classic integration-responsiveness framework, but it also shows that in the current geopolitical environment, the balance between integration and responsiveness must be managed as a continuous adaptive capability rather than as a stable strategic posture.

8.4 Dynamic Capabilities, Horizon Scanning, and Policy Alignment

If business model adaptation is understood as reconfiguration, then dynamic capabilities are the organisational processes that make reconfiguration possible. Teece (2007) identifies three such capabilities: sensing environmental shifts, seizing opportunities, and transforming internal structures. The interview data suggests that Danish life science companies have developed relatively strong sensing capabilities. Managers were acutely aware of competitive threats from Chinese generics, policy directions in the 15th Five-Year Plan, and geopolitical risks affecting regulatory recognition and market access.

However, seizing and transforming capabilities appeared weaker. Headquarters response lagged behind local perception, talent pipelines were constrained, and governance routines created friction. This imbalance between sensing and acting is a central finding of the study. It implies that dynamic capabilities should not be treated as a uniform competence, in multinational firms, they may be distributed unevenly across geographic levels, with subsidiaries stronger at sensing and headquarters stronger at formal resource allocation but slower at transforming.

The Three Horizons framework (Cuhls, 2020; Wintle et al., 2020) helps structure this temporal dimension of adaptation. Horizon 1 activities focused on defending existing revenue streams and operational efficiency, such as managing patent expiries and cost competition. Horizon 2 involved transitional practices, including B2C digital channels, regional innovation models, and local R&D expansion. Horizon 3 concerned long-term positioning, particularly alignment with China's 15th Five-Year Plan priorities in life science, sustainability, and advanced manufacturing.

This alignment with Chinese industrial policy is not simply a reactive compliance strategy; it is a form of forward-looking positioning that mirrors the Chinese government's own long-term planning logic. The interview material suggests that when firms align their investments with policy priorities such as green transition, scope 3 emissions reduction, and life science modernisation they are more likely to secure government support and market access. In this sense, Chinese five-year and fifteen-year planning can be read as a horizon-scanning tool that creates predictability within uncertainty. Firms that understand this logic and build relationships with central and municipal authorities are better positioned to navigate regulatory complexity than firms that rely solely on home-country government channels.

8.5 Risk, Resilience, and the Logic of Diversification

The DBOC curve from 2021 to 2025 documents a broader shift from market optimism to risk-aware engagement. This shift is consistent with Christopher and Peck's (2004) argument that supply chains designed primarily for efficiency become vulnerable when unexpected disruptions occur. It also resonates with Ivanov and Dolgui's (2020) emphasis on the survivability of intertwined supply networks, where disruption in one node can cascade through the system.

The interviewed firms responded to this risk environment through diversification rather than exit. Novonesis explicitly framed its strategy as not putting "all your eggs into one basket," maintaining a broad industry portfolio and expanding from one to two production sites in China. Novo Nordisk similarly emphasised global diversification of R&D and production as a competitive necessity. This suggests that de-risking, in the sense used by European policymakers, does not necessarily mean disengagement from China. Rather, it means reducing dependence on any single geography, supplier, or activity while continuing to participate in the Chinese market.

The finding also speaks to Courtney, Kirkland, and Viguerie's (1997) levels of uncertainty. The China environment during the study period often reflected Level 3 or Level 4 uncertainty, in which multiple possible futures coexist and the range of outcomes is not fully knowable. Under such conditions, optionality and flexibility become more valuable than optimisation. Maintaining parallel innovation tracks, local production capacity, and diversified sourcing creates strategic options that can be exercised as the future becomes clearer. Resilience, therefore, is not a property of any single business model component; it is an emergent property of the activity system as a whole.

8.6 Talent, Culture, and the Reduction of Decoupling

An unexpected but important theme emerging from the interviews is what we in our working notes termed the "people problem": the risk that deep tacit knowledge will leave the workforce as experienced employees retire, and that qualified replacements may not be available. This issue is not prominent in the traditional BMI literature, which tends to focus on strategic choices and structural redesign. Yet the data show that human capital is a critical mediator of business model reconfiguration. Without people who can bridge headquarters and subsidiary logics, localisation of R&D and manufacturing cannot proceed effectively.

The decline in expatriate managers with bicultural experience has created a "cross-cultural management gap," reducing the number of natural boundary spanners between Denmark and China. At the same time, localising management to Chinese nationals introduces new recruitment and retention challenges, particularly when technical infrastructure is concentrated in cities that may not match labour-market preferences. These findings suggest that talent strategy should be treated as an integral part of business model adaptation, not as an afterthought.

The working notes observation that "culture development (alignment) in a firm … narrows down decoupling" finds empirical support in the emphasis that interviewed firms placed on shared values and communication loops. Novonesis's use of global culture elements of: explore, ownership, care, and impact, is one example of an attempt to align behaviour across regions. However, the same manager acknowledged that not all headquarters colleagues adapted easily to regional practices, and that repeated communication was needed to reach agreement. This confirms that culture and communication are not substitutes for structural delegation, but complements to it. Information asymmetry narrows when both sides invest in understanding the other's institutional environment: why approvals take longer in one country, why certain documentation is required, or why a globally standard process may not fit local conditions.

8.7 Practical Implications for Danish Life Science Companies in China

The findings of this study carry several practical implications for Danish life science companies operating in or considering China. These implications are particularly relevant for firms with Danish

headquarters and China-based subsidiaries, though many are generalisable to other knowledge-intensive multinational sectors.

First, firms should treat government and stakeholder relationships as a strategic capability, not merely as a compliance function. The Chinese institutional environment rewards long-term alignment with policy direction and sustained engagement with central and municipal authorities. Companies that understand the logic of five-year and fifteen-year planning, and that position their investments accordingly, are more likely to secure predictable market access than those that react to policy announcements on a case-by-case basis.

Second, diversification should be understood as a structural principle rather than a short-term reaction to a crisis. This includes geographic diversification of production and R&D, supplier diversification, and portfolio diversification across products and customer segments. The goal is not to eliminate China exposure, the life science demand in China remains large and long-term, but to avoid over-dependence on any single node in the activity system.

Third, companies should invest deliberately in talent pipelines and knowledge transfer mechanisms. The departure of experienced staff, whether expatriate or local, threatens the tacit knowledge that underpins effective HQ-subsidiary coordination. Formal mentoring, rotation programmes, and documentation of institutional knowledge can reduce this risk. Equally important is the cultivation of bicultural managers who can translate between headquarters and local institutional logics.

Fourth, reducing decoupling requires both structural empowerment and cultural alignment. Subsidiaries need genuine decision authority over time-sensitive local activities, bounded by clear strategic guardrails. At the same time, headquarters and subsidiaries need sustained communication routines that build mutual understanding of each other's constraints. As the working notes emphasise, "information is power": the more each side understands the other's operating environment, the less scope there is for irritation, delay, and misalignment.

8.8 Contribution to Business Model Innovation Research

This study makes two main contributions to the BMI literature. First, it extends the activity system perspective by showing how adding, linking, and reallocating activities unfold in a high-uncertainty, institutionally distant host market. While Zott and Amit (2010) provide the conceptual vocabulary for these mechanisms, this study offers an empirical account of how they interact over time and how they are constrained by headquarters-subsidiary dynamics and talent availability.

This contribution responds directly to the gap identified in Section 2.4 by showing not only that business model thinking changes under simultaneous shocks, but which activity mechanisms and business model components become central when resilience, geopolitical uncertainty, and decoupling occur together.

Second, the study contributes to the growing literature on BMI under uncertainty by connecting macro-level environmental shocks with micro-level organisational adaptation. Foss and Saebi (2017) call for more research on the triggers and conditions of business model change. This study responds by identifying geopolitical tension, regulatory divergence, supply-chain fragility, and demand weakness as interconnected triggers, and by showing that their impact is mediated by internal coordination structures. The implication is that future BMI research should pay closer attention to the organisational mechanisms through which external shocks are translated or fail to be translated into business model change.

This also completes the gap-finding-contribution chain by showing that macro-level shocks affect BMI through concrete organisational mechanisms: headquarters-subsidiary coordination, subsidiary sensing, talent availability, local partnerships, and the reallocation of activities across borders.

Finally, the study bridges business model theory with international business research on headquarters-subsidiary relationships. By showing how decoupling and the dual business model trade-off shape BMI, it demonstrates that business model adaptation in multinational firms cannot be understood without reference to the distribution of power, knowledge, and legitimacy between headquarters and subsidiaries. This integration of perspectives is particularly needed as geopolitical fragmentation makes the internal organisation of multinational firms an increasingly important determinant of competitive success.

9. Conclusion

9.1 Answering the Research Question

This study set out to examine how Danish companies' perspectives on China evolved between 2021 and 2025, and how Danish companies in the life science sector can adapt their business models to build resilience against geopolitical uncertainty and organisational decoupling. Drawing on the DBOC sentiment index and qualitative interviews with industry representatives and corporate managers, the study traced a clear evolution in Danish business sentiment: from optimism in early 2021, through a severe downturn in 2022, to an uneven recovery and eventual stabilisation around neutral levels in 2024–2025.

The answer is developed through the DBOC dataset, institutional interviews, and empirical interviews with Novonesis and Novo Nordisk. The findings are derived from industry-level analysis supported by illustrative organisational insights, rather than from single-case generalisation.

The answer to the research question is that Danish life science companies in China have moved from a growth-oriented, headquarters-centred business model toward a more locally embedded, risk-aware, and adaptive model. This transition is characterised by the localisation of key activities, the reconfiguration of partnerships, the development of digital and direct-to-consumer channels, and the reallocation of human resources toward local talent. At the same time, adaptation remains uneven: organisational decoupling, talent constraints, and geopolitical ambiguity continue to limit how quickly and completely firms can reconfigure their activity systems.

9.2 Key Findings by Sub-Question

The first sub-question asked what overall trends and turning points could be identified in the DBOC data from 2021 to 2025. The analysis identified four phases: (1) early optimism in 2021, driven by China's role as a sales and production market; (2) a sharp crisis in 2022, triggered by COVID-19 lockdowns, supply-chain disruption, and the war in Ukraine; (3) an uneven recovery in 2023, with strong early momentum followed by renewed weakness; and (4) a phase of stabilisation and risk management in 2024–2025, in which firms adjusted to weaker demand, price competition, and persistent geopolitical tension.

The second sub-question asked what risks and challenges are most relevant for a Danish life science company with Danish headquarters and China-based operations. The most important risks are organisational decoupling between headquarters and subsidiary, regulatory and data-security complexity, supply-chain vulnerability, local cost competition, and the erosion of tacit knowledge as experienced staff retire. Geopolitical tension amplifies these risks by introducing uncertainty about market access, regulatory recognition, and the future viability of cross-border innovation partnerships.

The third sub-question asked how such a company can adapt its business model to balance resilience, local responsiveness, and continued China market engagement. The study identifies four adaptive moves: localising R&D and manufacturing to reduce trade and currency exposure; reconfiguring partnerships to include local-Chinese actors alongside global-Danish ones; developing digital and B2C channels to capture changing customer behaviour; and investing in local talent and cross-cultural alignment to narrow headquarters-subsidiary decoupling. These moves are most effective when pursued as interdependent elements of a broader activity system, supported by dynamic capabilities and guided by horizon-scanning processes.

9.3 Implications for Practice

For practice, the study suggests that Danish life science companies should move beyond reactive crisis management and develop deliberate adaptation capabilities. This includes building long-term government relationships in China, diversifying geographic and operational exposure, empowering local subsidiaries while maintaining strategic coherence, and investing in talent and knowledge transfer. Resilience is not achieved through a single strategic decision but through the cumulative development of organisational capabilities that enable ongoing reconfiguration.

9.4 Limitations and Directions for Future Research

This study was deliberately scoped around a small, purposive interview sample, empirical company interviews, and cross-sectionally collected qualitative data. These design choices prioritised analytical depth within a specific institutional context over broad generalisability. The DBOC sample is restricted to member companies of DCCC and DCBF and measures sentiment rather than objective performance, while the case focus on large Danish life science companies limits the extent to which findings can be transferred directly to small and medium-sized enterprises. The complexity of the business environment and regulations surrounding the life science industry therefore makes the generalizability of this study to other sectors more complex, whereas some points can be generalised to other industries, other points made in this study will not.

Future research could extend this study in several directions. Comparative studies across sectors and nationalities would clarify which findings are specific to life science, such as regulatory intensity, patient access, patent expiry, clinical trials, and price-volume tradeoffs, and which may travel more easily to other industries, such as headquarters-subsidiary decoupling, local sensing, diversification, and talent/knowledge-transfer issues. Longitudinal case studies could trace how individual companies reconfigure their activity systems over time and how the headquarters-subsidiary balance evolves. Finally, research could examine more closely the micro-foundations of dynamic capabilities in multinational firms, particularly the mechanisms through which subsidiary-level sensing is translated into headquarters-level action.

In conclusion, the period 2021–2025 challenged many of the assumptions that had long underpinned Danish life science companies China strategies. Yet it also revealed that adaptation is possible when firms treat their China operations not as a distant market outlet but as an integrated, learning-oriented part of a globally diversified activity system. The task ahead is not to predict the future of China, but to build the organisational capabilities necessary to adapt continuously as that future unfolds.

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Appendices

Appendix A: Scopus Search Log and Search Strategy

This appendix documents the Scopus search strategy used to support the literature review and theoretical framework. It records what was searched, why the searches were relevant, and how sources were selected or excluded. The purpose is to make the literature collection process transparent, traceable, and easy to replicate.

A.1 Search scope

Element	Scope used in the project
Database	Scopus
Search fields	Article title, abstract, and keywords: TITLE-ABS-KEY(...)
Document types	Articles, review articles, conference papers where relevant, and book chapters only where theory-defining sources are needed
Language	English
Publication years	1983–2026 for foundational theory; 2010–2026 for current business model, geopolitical risk, resilience, and China/MNE literature
Reason for mixed date range	Some core theories are older but still central, especially institutional theory, business model theory, headquarters-subsidiary theory, and dynamic capabilities. A recent-only search would exclude these foundational sources.

A.2 Core search concepts

The search was structured around five literature clusters and a narrower empirical context. Because the exact niche of Danish life science companies in China is specific, the search was designed in layers: broad theory searches first, then contextual narrowing searches.

Literature clusters:

- Business models and business model innovation
- Dynamic capabilities, uncertainty, and strategic adaptation
- Institutional legitimacy and headquarters-subsidiary coordination
- Supply-chain resilience and geopolitical risk
- Horizon scanning and foresight

Empirical context:

- Danish companies in China
- Multinational healthcare/life-science companies in China
- China market uncertainty, regulation, and de-risking
- DBOC sentiment data as the empirical background for Danish business perspectives

A.3 Search log

The table below records the main thematic Scopus searches. The count columns are kept as completion fields, since the final result counts should be entered after running each string directly in Scopus.

ID	Search focus	Filters	Results/screening status	Reason for use
S1	String A: Broad business model adaptation under uncertainty	English; article/review; 1983–2026	Results / screened / selected: to be filled after running search	Broad search for BMI, uncertainty, and China/MNE context
S2	String B: Business model innovation and dynamic capabilities	English; article/review; 2007–2026	Results / screened / selected: to be filled after running search	Dynamic capabilities and strategic adaptation
S3	String C: Business Model Canvas / activity-system perspective	English; article/review/book chapter; 2002–2026	Results / screened / selected: to be filled after running search	Business Model Canvas and activity-system framework
S4	String D: Headquarters-subsidiary coordination in China	English; article/review; 1987–2026	Results / screened / selected: to be filled after running search	HQ-subsidiary coordination in China/MNEs
S5	String E: Institutional legitimacy and multinational strategy	English; article/review; 1983–2026	Results / screened / selected: to be filled after running search	Institutional legitimacy and institutional pressures
S6	String F: Supply-chain resilience and geopolitical risk	English; article/review; 2004–2026	Results / screened / selected: to be filled after running search	Supply-chain resilience and geopolitical risk
S7	String G: Horizon scanning and foresight	English; article/review; 2015–2026	Results / screened / selected: to be filled after running search	Horizon scanning and weak signals
S8	String H: Healthcare / life-science multinationals in China	English; article/review; 2010–2026	Results / screened / selected: to be filled after running search	Healthcare/life-science multinationals in China
S9	String I: Danish companies in China	English; article/review; 2010–2026	Results / screened / selected: to be filled after running search	Contextual check for Danish companies in China
S10	String J: China de-risking and decoupling	English; article/review; 2018–2026	Results / screened / selected: to be filled after running search	China de-risking, decoupling, and trade tensions

A.4 Master Scopus search strings

String A: Broad Business Model Adaptation Under Uncertainty

```
TITLE-ABS-KEY(
 ("business model*" OR "business model innovation" OR "business model adaptation" OR "business model reconfiguration" OR "business model design")
 AND
 (uncertain* OR disruption* OR "external shock*" OR "environmental dynamism" OR "geopolitical risk" OR "institutional environment*" OR resilience)
 AND
 (China OR Chinese OR "emerging market*" OR multinational* OR MNE OR subsidiary OR subsidiaries)
)
```

Use: Main entry search for literature connecting business model change with uncertainty and China/MNE contexts.

String B: Business Model Innovation And Dynamic Capabilities

```
TITLE-ABS-KEY(
 ("business model innovation" OR "business model adaptation" OR "business model reconfiguration")
 AND
 ("dynamic capabilities" OR "sensing" OR "seizing" OR transforming OR reconfiguration)
 AND
 (uncertain* OR disruption* OR "environmental change" OR "strategic adaptation")
)
```

Use: Supports the theoretical link between Teece, business model innovation, and adaptive response.

String C: Business Model Canvas / Activity System Perspective

```
TITLE-ABS-KEY(
 ("business model canvas" OR "business model design" OR "activity system" OR "value creation" OR "value capture")
 AND
 ("business model*" OR firm* OR organization* OR organisation*)
)
```

Use: Supports the analytical framework, especially the use of BMC as a structure for categorising business model components.

String D: Headquarters-Subsidiary Coordination In China

```
TITLE-ABS-KEY(
 ("headquarters-subsidiary" OR "headquarter-subsidiary" OR "HQ-subsidiary" OR subsidiary OR subsidiaries)
 AND
 ("local responsiveness" OR "global integration" OR coordination OR control OR legitimacy)
 AND
 (China OR Chinese OR "emerging market*")
 AND
 (multinational* OR MNE OR "foreign firm*" OR "foreign subsidiary")
)
```

Use: Finds literature explaining the gap between Denmark-based and China-based perspectives.

String E: Institutional Legitimacy And Multinational Strategy

```
TITLE-ABS-KEY(
 ("institutional legitimacy" OR "institutional theory" OR "institutional distance" OR "institutional environment*")
 AND
 (multinational* OR MNE OR subsidiary OR subsidiaries)
 AND
 (China OR Chinese OR "emerging market*")
)
```

Use: Supports legitimacy, local adaptation, and institutional pressure arguments.

String F: Supply-Chain Resilience And Geopolitical Risk

```
TITLE-ABS-KEY(
 ("supply chain resilience" OR "supply chain risk" OR "supply chain disruption*" OR "supply network*" OR survivability)
 AND
 ("geopolitical risk" OR geopolitic* OR "trade tension*" OR decoupling OR "de-risking" OR sanction* OR tariff*)
 AND
 (China OR Chinese OR global OR multinational* OR MNE)
)
```

Use: Connects operational continuity, resilience, and geopolitical uncertainty.

String G: Horizon Scanning And Foresight

```
TITLE-ABS-KEY(
 ("horizon scanning" OR foresight OR "strategic foresight" OR "weak signal*" OR "early warning")
 AND
 (uncertain* OR disruption* OR "emerging issue*" OR "strategic planning" OR preparedness)
)
```

Use: Supports the paper's forward-looking method for interpreting emerging risks and future signals.

String H: Healthcare / Life Science Multinationals In China

```
TITLE-ABS-KEY(
 (healthcare OR "health care" OR pharmaceutical* OR "life science*" OR biotech* OR medtech OR "medical device*")
 AND
 (China OR Chinese)
 AND
 (multinational* OR MNE OR "foreign firm*" OR subsidiary OR subsidiaries)
 AND
 (regulation OR "market access" OR localization OR localisation OR "supply chain" OR innovation OR uncertainty)
)
```

Use: Narrows the literature toward the sector context of the case company.

String I: Danish Companies In China

```
TITLE-ABS-KEY(
 (Danish OR Denmark)
 AND
 (China OR Chinese)
 AND
 (business OR firm* OR compan* OR multinational* OR subsidiary OR subsidiaries)
)
```

Use: Contextual search. Expected to produce fewer results, but useful for documenting that the exact country-company context was checked.

String J: China De-Risking And Decoupling

```
TITLE-ABS-KEY(
 (China OR Chinese)
 AND
 ("de-risking" OR decoupling OR "geopolitical risk" OR "trade tension*" OR "strategic autonomy" OR "supply chain diversification")
 AND
 (business OR firm* OR multinational* OR investment OR "foreign direct investment" OR FDI)
)
```

Use: Supports the contextual argument that China engagement has shifted from growth optimism toward risk management.

A.5 Seed article recovery searches

These searches were included to show that the project did not only use broad keyword searches. They also verify key seed articles already used in the literature review and theoretical framework. For each source, Scopus indexing, citation count, and retention status should be recorded after searching.

Literature cluster	Seed article recovery string	Action/status
Business models and BMI	TITLE("Business models, business strategy and innovation") AND AUTHOR-NAME(Teece)	Verify Scopus record/citations; retain if indexed
Business models and BMI	TITLE("The business model: Recent developments and future research") AND AUTHOR-NAME(Zott)	Verify Scopus record/citations; retain
Business models and BMI	TITLE("Business model innovation: Opportunities and barriers") AND AUTHOR-NAME(Chesbrough)	Verify Scopus record/citations; retain if relevant
Business models and BMI	TITLE("Fifteen years of research on business model innovation") AND AUTHOR-NAME(Foss)	Verify Scopus record/citations; retain
Business models and BMI	TITLE("Business model design: An activity system perspective") AND AUTHOR-NAME(Zott)	Verify Scopus record/citations; retain
Business models and BMI	TITLE("Managing the dual business model trade-off in multinational corporations") AND AUTHOR-NAME(Mozheiko)	High-priority source; verify indexing

Business models and BMI	TITLE("Dual performance of business model innovation in emerging market enterprises") AND AUTHOR-NAME(Huang)	Verify and complete reference if retained
Dynamic capabilities and uncertainty	TITLE("Explicating dynamic capabilities") AND AUTHOR-NAME(Teece)	Verify Scopus record/citations; retain
Dynamic capabilities and uncertainty	TITLE("Strategy under uncertainty") AND AUTHOR-NAME(Courtney)	Verify source type; may be HBR/contextual theory
Dynamic capabilities and uncertainty	TITLE("Why business models matter") AND AUTHOR-NAME(Magretta)	Verify source type; may be HBR/contextual theory
Dynamic capabilities and uncertainty	TITLE("Reinventing your business model") AND AUTHOR-NAME(Johnson)	Verify source type; may be HBR/contextual theory
Institutional and HQ-subsidiary	TITLE("The iron cage revisited") AND AUTHOR-NAME(DiMaggio)	Verify Scopus record/citations; foundational source
Institutional and HQ-subsidiary	TITLE("Organizational legitimacy under conditions of complexity") AND AUTHOR-NAME(Kostova)	Verify Scopus record/citations; retain
Institutional and HQ-subsidiary	TITLE("Adoption of an organizational practice by subsidiaries of multinational corporations") AND AUTHOR-NAME(Kostova)	Verify Scopus record/citations; retain
Institutional and HQ-subsidiary	TITLE("Headquarters-subsidiary relationships in MNCs") AND AUTHOR-NAME(Kostova)	Verify Scopus record/citations; retain if used
Institutional and HQ-subsidiary	TITLE("Evolution of MNE strategies amid China's changing institutions") AND AUTHOR-NAME(Tse)	Verify Scopus record/citations; retain if used
Supply chain and geopolitical risk	TITLE("Building the resilient supply chain") AND AUTHOR-NAME(Christopher)	Verify Scopus record/citations; retain
Supply chain and geopolitical risk	TITLE("Viability of intertwined supply networks") AND AUTHOR-NAME(Ivanov)	Verify Scopus record/citations; retain
Supply chain and geopolitical risk	TITLE("Measuring geopolitical risk") AND AUTHOR-NAME(Caldara)	Verify Scopus record/citations; retain
Horizon scanning and foresight	TITLE("Horizon scanning in foresight") AND AUTHOR-NAME(Cuhls)	Verify exact source and reference details
Horizon scanning and foresight	TITLE("Scanning the horizon: a systematic literature review of methodologies") AND AUTHOR-NAME(Hines)	Verify Scopus record and add full reference if retained
Horizon scanning and foresight	TITLE("Bioengineering horizon scan 2020") AND AUTHOR-NAME(Wintle)	Verify Scopus record and add full reference if retained

A.6 Inclusion and exclusion criteria

Inclusion criteria:

- The source offered a foundational or highly cited concept used in the analytical framework.
- The source addressed business model innovation, adaptation, or reconfiguration under uncertainty.

- The source discussed multinational strategy, institutional legitimacy, or headquarters-subsidiary coordination.
- The source examined supply-chain resilience, geopolitical risk, de-risking, or China-related strategic uncertainty.
- The source provided methodological support for horizon scanning, foresight, or weak-signal interpretation.
- The source was directly relevant to life science, life sciences, or foreign firms operating in China.

Exclusion criteria:

- The source was not available in English.
- The source focused only on consumer behaviour or technology adoption without business model or strategic relevance.
- The source addressed China only as a generic market without institutional, geopolitical, regulatory, or multinational relevance.
- The source was not peer-reviewed, unless it was used as contextual data rather than academic literature.
- The source was a duplicate record or an older version of the same study.
- The source did not contribute directly to the research question or sub-questions.

A.7 Quality and ethicality note

The Scopus search was used to improve transparency, traceability, and quality in the literature review. Search strings, database choice, filters, and screening criteria were documented so that the search process could be repeated or challenged. Foundational sources were retained even when older than the main contemporary date range because they define central concepts used in the study. Newer sources were prioritised where the purpose was to understand contemporary uncertainty, geopolitical risk, business model adaptation, and China-related multinational strategy.

Non-academic sources such as DBOC reports, government publications, World Bank reports, Reuters, MERICS, and policy documents should be treated as contextual or empirical sources rather than as part of the peer-reviewed Scopus literature base. This distinction supports research integrity by making clear which sources are academic literature and which sources describe market, regulatory, policy, or historical developments.

Appendix B- Data Profile

Gioia Coding Structure and Supporting Interview Evidence

This appendix supports the condensed interview findings presented in Chapter 6. It shows how the interview material was organised using a Gioia-based coding logic: first-order concepts were kept close to interviewee language, grouped into second-order themes, and then aggregated into broader dimensions used in the analysis.

The appendix is not intended to replace the full interview coding database. Instead, it provides a transparent bridge between the coded interview excerpts and the shorter findings section in the main report.

B.1 Gioia Coding Structure

Aggregate dimension	Second-order themes	First-order concepts and illustrative evidence
China as a strategically necessary but more contested market	Market scale and strategic importance; China as innovation and R&D system; stronger domestic competition; China policy speed and implementation intensity.	China described as Novo Nordisk's second biggest affiliate after the US; China viewed as a major R&D centre; Chinese firms developing stronger original molecules; institutional interviewees describe China as a market Danish firms cannot ignore because competition can travel globally.
Localised business model reconfiguration	Local R&D and production; China-specific application development; digital/direct channels; local partnerships and licensing; activity relocation from HQ/global systems into the China organisation.	Novo Nordisk describes a China 'closed loop model' with R&D, production, and commercial activity in China; Novonesis describes regional innovation and production localisation; DCCC observes e-commerce and Chinese platform partnerships.
Headquarters-subsidiary decoupling and decision friction	Information lag between Denmark and China; communication costs; negotiated autonomy; HQ risk perception; local managers needing faster authority.	DCCC describes Denmark-based sentiment as one to two quarters slower than China-based reality; Novonesis describes repeated communication loops and power bargaining between region and headquarters; Novo Nordisk describes shifting affiliate freedom and HQ control.
Risk-aware commitment rather than withdrawal	De-risking as diversification; staying in China to learn from competition; global portfolio risk across US, China, and Europe; long-term China commitment despite uncertainty.	Novonesis rejects 'escaping' from China and argues that staying helps the company learn from Chinese players; Novo Nordisk links China resources to diversification; DCBF describes simultaneous pressure from the US on pricing/revenue and from China on competition/market share.
Healthcare/life-science specific adaptation pressures	Patent expiry and price-volume trade-offs; patient access; regulatory/data compliance; clinical development costs; health-policy alignment.	Novo Nordisk frames patent expiry as normal pharma lifecycle management and describes the volume-price trade-off; DCCC highlights rigorous medical testing governance, IP/data concerns, and cheaper clinical trials in China; interviewees connect health demand with public-policy priorities.
Talent, tacit knowledge, and cultural translation	Local leadership; declining expatriate boundary-spanning; talent geography; tacit market understanding; culture as coordination mechanism.	DCCC describes fewer young Danish managers in China and stresses that 'seeing is understanding'; Novonesis identifies difficulty attracting high-quality talent when labs and talent pools are geographically separated; DCCC notes experienced Chinese middle managers inside Danish firms.

B.2 Selected Supporting Interview Evidence

Finding/theme	Participant	Supporting quote
China remains strategically necessary	Participant 1 - Novo Nordisk	"We operate here for more than 20 years... from Novo Nordisk's perspective, it is the second biggest affiliate in the world, right after the USA."
China as an innovation partner ecosystem	Participant 2 - Novonesis	"In China we are able to find some more qualified partners to choose from. And that's the difference compared to maybe 10 or decades ago."
Localised business model reconfiguration	Participant 1 - Novo Nordisk	"We have R&D also in China, we have production also in China, and these affiliate or commercial parts also in China."
Regional innovation/application development	Participant 2 - Novonesis	"The scope of regional innovation is not to develop or invent a new backbone of the enzyme. It is to take enzyme products off the shelf and develop new applications to serve regional customer needs."
HQ-China sentiment lag	Participant 3 - DCCC	"The sentiment in Denmark is slower to react to the reality on the ground in China... it takes about one to two quarters for it to reflect in the data in Denmark."
Need for local empowerment	Participant 2 - Novonesis	"We need to be empowered even more, and we need to reduce the communication costs between the region and the global organization."
De-risking as staying and learning	Participant 2 - Novonesis	"Because the local competition is growing, we should stay here rather than escaping from here. We need to be part of this society and really follow what's going on here."
Dual US-China pressure	Participant 4 - DCBF	"Danish companies now face pressure from both sides: from the US on pricing and revenue, and from China on competition and market share."
Healthcare price-volume trade-off	Participant 1 - Novo Nordisk	"Now is the game... how much volume versus how much price... price went down, we are increasing volume."
Healthcare regulatory intensity	Participant 3 - DCCC	"China's governance of their medical testing and laboratory testing is quite rigorous, and you have to file a lot of applications."
Declining Danish boundary-spanners	Participant 3 - DCCC	"The number of young... Danish managers coming out to China with that cultural context of both China and Denmark has decreased significantly."
Talent geography as constraint	Participant 2 - Novonesis	"I found it more and more difficult to attract very high quality talents to our company... I would like to hire that person in Beijing, sitting close to my lab."

B.3 DBOC Data Profile

This chapter presents the quantitative foundation for the study. Its purpose is to set out the data landscape within which the analysis in subsequent chapters takes place. By describing the DBOC Index dataset, justifying the focus on the life science sector, and mapping the five-year sentiment trajectory in detail, the chapter establishes the analytical ground on which questions of business model adaptation, organisational decoupling, and strategic resilience will be examined.

B.3.1 Dataset Overview

The Danish Business Outlook on China (DBOC) Index is a quarterly business sentiment survey jointly developed by the Danish Chamber of Commerce in China (DCCC), the Danish-Chinese Business Forum (DCBF), the Embassy of Denmark in Beijing, and the Sino-Danish Center (SDC). The index tracks Danish companies' short-term outlook on China across four business functions,which are Sales, Manufacturing/Production, Procurement, and Research & Development (R&D) ,scored on a 0-to-100 scale where 50 represents neutral sentiment. Respondents are classified across six sectors,which are Energy, Health / Life Science, Agriculture / Food, Environment / Water, Logistics / Maritime, and Design / Lifestyle. The dataset covers the period from Q1 2021 to Q4 2025, comprising 20 survey rounds and capturing Danish firms' sentiment evolution through a sequence of major external disruptions including COVID-19 lockdowns, supply chain crises, and geopolitical tensions.

B.3.2 Healthcare and Life-Science Sector

For this study, the analysis is narrowed to the Health / Life Science sector. This sector accounts for 9 respondents in the 2025 Q4 survey round, representing 24.3% of the 37 sector-identified participants and making it the largest single sector in the dataset alongside Agriculture / Food and Logistics / Maritime. The healthcare subsample is dominated by large enterprises,87.5% of respondents are large firms with more than 250 employees globally, while only 12.5% are SMEs. Geographically, two-thirds of life science respondents are based in China, with the remaining third operating from Denmark.

The decision to focus on life science reflects its strategic significance within the Danish Trade Council's sector prioritisation in China. life science firms operate at the intersection of commercial activity, public policy, and regulation, making them especially sensitive to the geopolitical and operational developments that the DBOC dataset captures. Moreover, life science demand in China is relatively stable and long-term in nature, limiting firms' ability to withdraw quickly from the market. This combination of strategic importance, regulatory exposure, and market stickiness makes the life science sector a particularly instructive case for analysing how Danish companies adapt their business models under sustained uncertainty.

B.3.3 Contextual Setting

The external developments discussed in this section provide the contextual backdrop against which the DBOC data is interpreted. Rather than serving as theoretical explanations, these events function as the empirical conditions that shaped Danish companies' China outlook during the 2021-2025 period. The DBOC data and subsequent analysis examine how these contextual factors are reflected in company sentiment and business model considerations.

B.3.3.1 COVID-19 Lockdowns and Travel Restrictions

COVID-19 restrictions were one of the most important disruptions for companies in China during the period. China used strict lockdowns, quarantine rules, and travel restrictions for a longer time than many other countries. In 2022, the World Bank reported that China's economy was strongly affected by COVID-19 outbreaks and lockdowns, especially from March to May 2022 (World Bank, 2022a).

For companies, lockdowns and travel restrictions could weaken market access and delay projects. When managers, technical experts, or headquarters staff cannot travel, it becomes harder to visit customers, inspect operations, develop partnerships, or make investment decisions. This is especially relevant for international companies because many decisions depend on communication between headquarters and local subsidiaries.

From a business model perspective, COVID-19 restrictions affected value delivery and coordination. Companies had to rely more on local teams, digital communication, and remote management. In some cases, this may have increased the need for local autonomy, because headquarters could not be physically present in the market.

B.3.3.2 Supply-Chain Disruption

Supply-chain disruption became a major global issue during and after the COVID-19 pandemic. Lockdowns, port delays, transport shortages, and changing demand patterns made global supply chains less predictable. The World Bank (2022c) described how lockdowns in China and global supply-chain problems contributed to wider economic uncertainty.

This matters for business models because sourcing and production are part of how companies deliver value. If a company depends too much on one country, one supplier, or one logistics route, it becomes more vulnerable when disruption happens. Companies may therefore need to diversify suppliers, hold more inventory, regionalize sourcing, or create backup production options.

Christopher and Peck (2004) argue that supply chains designed mainly for efficiency can become vulnerable when unexpected disruptions happen. Ivanov and Dolgui (2020) also argue that COVID-19 showed the importance of looking at the whole supply network, not only individual suppliers. In practice, resilience means that companies try to build systems that can continue operating even when the external environment changes.

B.3.3.3 Ukraine War, Energy Shocks, and Cost Pressure

The Russian invasion of Ukraine in February 2022 created major uncertainty in the global economy. It affected energy prices, food prices, logistics, inflation, and overall business confidence. The World Bank (2022c) warned that the war in Ukraine, together with lockdowns in China and supply-chain disruption, created risks of global slowdown and stagflation.

For companies, this type of shock increases uncertainty and cost pressure. Higher energy and transport costs can affect production, sourcing, and pricing. Even when companies are not directly active in Ukraine or Russia, the war can still influence their global operations through inflation, financial uncertainty, and supply-chain effects.

From a business model perspective, this creates more attention to resilience and cost control. Companies may need to reconsider where they produce, how they price products, and how much risk they can accept in their supply chain. For life science companies, this can also affect input costs, distribution, and long-term investment timing.

B.3.3.4 China Reopening and Temporary Market Optimism

At the end of 2022 and beginning of 2023, China started to reopen after the zero-COVID period. The World Bank expected China's economy to recover in 2023, but also warned that risks remained, including property-sector stress, weak global demand, and geopolitical tensions (World Bank, 2022b).

For companies, reopening created new possibilities for travel, customer contact, market visits, and investment discussions. It also created expectations that Chinese demand would recover. From a business model perspective, this could support renewed interest in China as a sales market and investment location.

However, reopening did not remove all uncertainty. Companies still had to consider whether the recovery would be strong enough and whether the external environment had changed permanently. Therefore, reopening can be understood as a period of cautious re-engagement rather than a simple return to the pre-pandemic business environment.

B.3.3.5 Weak Chinese Demand and Property-Sector Crisis

After reopening, China's economic recovery was weaker than many expected. The World Bank reported that China's growth was affected by weak domestic demand and the property-sector downturn (World Bank, 2024). In 2025, the World Bank also highlighted soft consumption and uncertainty related to global trade policy (World Bank, 2025).

Chinese sources describe the same period more from a policy-stabilization perspective. The People's Bank of China stated that 2024 policy focused on stable growth, financial support for the real economy, and counter-cyclical adjustment (People's Bank of China, 2025). This shows that Chinese authorities were also aware of pressure in the economy, but framed the response around policy support and stability.

Weak demand matters for companies because it puts pressure on revenue models and growth assumptions. If companies entered China expecting fast market growth, slower demand may require them to adjust pricing, customer segments, sales channels, or investment plans.

The property-sector crisis is also important because property has been a major part of China's economy. When the sector slows down, it can affect consumer confidence, local government finances, investment, and wider economic expectations. For foreign companies, this means that China may still offer opportunities, but growth becomes more selective and less automatic.

B.3.3.6 Data and Security Regulation

China's regulatory environment also became more important during this period. The Data Security Law came into effect on September 1, 2021, and the Personal Information Protection Law came into effect on November 1, 2021 (National People's Congress, 2021a, 2021b). These official Chinese legal sources show that data governance, cybersecurity, personal information protection, and cross-border data transfer became more important for companies operating in China.

The revised Counter-Espionage Law also increased attention to national security and information risks. The Library of Congress (2023) explains that the revised law expanded the scope of espionage-related activities, including some documents, data, materials, and cyber activities connected to national security.

Chinese institutions also show that data governance is a strategic policy area. For example, Tsinghua University's data governance institute focuses on data governance, digital strategy, institutional systems, and national governance modernization (Tsinghua University, 2024). This supports the idea that data is not only a technical issue in China, but also a policy and governance issue.

For companies, this can make digital operations more compliance-heavy. It can also create caution around data sharing, analytics, R&D collaboration, due diligence, and communication between China and headquarters. This is especially relevant for life science companies because they may work with sensitive health data, clinical information, digital platforms, and regulatory documentation.

From a business model perspective, regulation can affect both value creation and value delivery. If data cannot move freely between China and headquarters, companies may need more local data infrastructure, local compliance capabilities, or China-specific digital processes.

B.3.3.7 Geopolitical De-Risking

Geopolitical tension has become an important factor in business decisions related to China. In Europe, the debate has increasingly moved from "decoupling" to "de-risking." This means that companies and governments do not necessarily want to leave China, but they want to reduce strategic dependencies and manage risks better. MERICS (2023) summarizes Ursula von der Leyen's position that Europe should not fully decouple from China, but should reduce risks in sensitive areas.

At the same time, Chinese official sources continue to present China as open to foreign investment. For example, the State Council Information Office reported that China introduced measures to expand market access, support fair competition, and facilitate innovation factors for foreign investors (State

Council Information Office, 2024). This shows an important contrast: European actors often talk about reducing dependency, while Chinese official sources emphasize opening-up and continued opportunities.

Geopolitical risk can reduce investment and increase uncertainty. Caldara and Iacoviello (2022) show that higher geopolitical risk is connected with lower investment and higher downside risks. For companies, this means that even attractive markets may become more difficult to invest in if the geopolitical environment is unstable.

From a business model perspective, de-risking can lead to less dependence on China-only production or China-only sourcing. Companies may keep China as an important market, but develop alternative suppliers, regional production options, or more flexible investment strategies.

B.3.3.8 China's 15th Five-Year Plan Priorities

China's upcoming 15th Five-Year Plan, covering 2026 to 2030, is relevant because Chinese five-year plans shape national policy priorities and business opportunities. The direction is expected to continue focusing on areas such as technological self-reliance, green transition, advanced manufacturing, life science, and high-quality growth.

For foreign companies, this can create opportunities if their business models align with Chinese policy priorities. Companies in life science, sustainability, advanced manufacturing, and technology may find new possibilities if they can connect their value proposition to China's long-term development goals.

Chinese official life science sources also show why life science is relevant. The State Council Information Office reported that China continued to deepen life science reform during the 14th Five-Year Plan period, including medical insurance coverage, affordable medicines, and support for innovative drugs (State Council Information Office, 2025). This matters for a Danish life science company because China may still offer opportunities if the company can align with local health policy priorities.

At the same time, policy alignment may require adaptation. Companies may need local partnerships, local R&D, local production, or China-specific solutions. This means that China-related business model innovation is not only about reducing risk, but also about finding where future opportunities fit with Chinese policy direction.

Appendix C: Dataset Yearly Overview

C.1 Year 2021

The DBOC reports for 2021 show that Danish companies' view of China changed quite a lot during the year. In the beginning of 2021, the outlook was clearly positive. In Q1, the index was 55.9, which shows that more companies experienced improvement than deterioration, and this was mainly because China was still seen as an important market for sales and exports. At this point, the general picture was optimistic, even though SMEs and respondents based in Denmark were already a bit less positive than larger companies and respondents based in China.

In Q2, the overall index stayed almost the same at 55.6, so on the surface the situation looked stable. However, the report also showed that the picture was becoming more uneven. Some sectors improved, while others became more negative, and travel restrictions into China started to affect many companies' short-term outlook.

In Q3, this weaker development became much clearer, as the index dropped to 48.7 and moved into negative territory. This suggests that Danish companies were now becoming more uncertain about their activities in China. The decline was mainly connected to weaker expectations regarding sales, manufacturing, and procurement, while R&D was the only area that did not worsen. At the same time, domestic travel restrictions inside China also had a negative effect, especially for SMEs and respondents based in China.

In Q4, the index increased slightly to 49.9, which shows a small improvement compared to Q3, but the outlook was still just below neutral. Sales, manufacturing, and R&D became more positive again, but procurement continued to weaken and remained the main factor pulling the overall outlook down.

Overall, the reports from 2021 suggest that Danish companies still saw China as an important market, especially in terms of sales, but during the year their outlook became more cautious and less stable because of operational difficulties, sector differences, and growing challenges related to procurement and business conditions.

C.2 Year 2022

The year 2022 represents the most severe period of negative sentiment in the entire DBOC dataset. The index had already begun declining through late 2021, but 2022 marked a qualitative shift in both the depth and nature of the challenges facing Danish companies in China.

The numbers tell the story plainly. Q1 registered 47.5, Q2 collapsed to 39.4, the lowest reading across the entire dataset, Q3 recovered marginally to 41.8, and Q4 closed at 39.9. Not a single quarter broke above the neutral threshold of 50.

Three external shocks converged simultaneously: prolonged COVID lockdowns including the Shanghai shutdown, severe supply chain disruption, and the Russian invasion of Ukraine. As Johnson, Christensen, and Kagermann (2008) observe, few companies understand their existing business model's interdependencies until those interdependencies are stressed by external disruption. 2022 was precisely that stress test.

The sales sub-category bore the heaviest damage, falling to a dataset low of 36.5 by Q4, having been the strongest dimension just eighteen months earlier. R&D alone held above 50 in Q1, hinting that firms oriented around knowledge-creation rather than market-access proved comparatively resilient.

A nine-point gap between China-based and Denmark-based respondents in Q2 further showed information asymmetry within Danish companies that would continue shaping strategic responses well beyond the year itself.

C.3 Year 2023

The year 2023 marked a transition from the sharp contraction of 2022 to an uneven and unstable recovery. The DBOC Index rebounded strongly in Q1 to 56.5, reflecting a rapid return of optimism, before increasing to 51.4 in Q2. However, this momentum weakened in the second half of the year, with Q3 slipping below the neutral threshold of 50 to 49.4 and Q4 declining further to 47.8. Despite the strong start, the recovery proved fragile and inconsistent.

The initial improvement in Q1 was broad-based, with all major market dimensions turning positive. Sales rebounded most significantly, once again acting as the primary driver of sentiment. Yet this strength was not sustained. By Q3, sales had turned negative again, pulling down the overall index and highlighting its high sensitivity to changing conditions. In contrast, manufacturing and procurement remained comparatively stable throughout the year, indicating slower and more gradual adjustment processes. A notable divergence emerged between short-term performance and longer-term strategic outlooks. While sales fluctuated sharply, R&D remained relatively weak and showed only limited improvement over the year. This suggests continued caution in long-term investment despite short-term optimism earlier in the year.

At the same time, the recovery became increasingly uneven across sectors and firm types. Early gains were widely shared, but by mid-year, performance diverged more clearly, with some industries maintaining stability while others declined. SMEs showed greater volatility, reacting more sharply to both improvements and downturns, while larger firms remained more stable but subdued. Differences between China-based and Denmark-based respondents narrowed compared to 2022 but persisted, reflecting ongoing variation in internal market perceptions.

C.4 Year 2024

Throughout 2024, DBOC Index stayed stable around the neutral 50 point level. This showed a clear strategic shift from the high recovery hopes seen in late 2023 to a more realistic phase of risk management.

In terms of the four market categories, the most notable feature of 2024 was the clear gap between short-term sales expectations and long-term research and development (R&D) investments. Sales remained the weakest sector all year, moving between 45 and 52 points, which represents a period of slow growth compared to the high-growth years like 2021 and early 2023. This weakness was mainly driven by the fierce price wars that intensified across Chinese industries in early 2024. For example, aggressive price cuts by domestic giants like BYD forced multinational companies to accept much lower profit margins just to stay competitive (Reuters, 2024). In contrast, the manufacturing index remained relatively stable, reflecting the steady commitment of Danish firms to their localization strategy,which helps companies serve the domestic market more efficiently and reduce supply chain risks. Meanwhile, the R&D index was the star performer, staying between 55 and 60 even when the sales index acted poorly. This shows a clear strategy that while current sales are difficult, investing in R&D is seen as the vital part to build technical advantages for the future.

The performance across different industries also revealed how business confidence was affected by government policy and trade issues. The industrial and manufacturing sectors performed better than others, supported by the Action Plan for Large-scale Equipment Renewal launched in March (State Council of the People's Republic of China, 2024). This government policy created a direct demand for Danish companies that sell energy-saving machines and industrial solutions. On the other hand, the food and consumer goods sectors faced a lot of instability, especially in the middle of the year. This downturn was directly linked to growing trade tensions between the EU and China. After the EU put taxes on Chinese electric cars, China started investigations into European pork and dairy imports (Ministry of Commerce of the People's Republic of China, 2024). Since Denmark exports a lot of these products, these trade disputes brought high political risks and hurt sales forecasts for those firms.

Finally, the data regarding geographic location and company size shows how different organizations are adapting. Trends in 2024 revealed a clear difference in outlook between Danish headquarters and managers on the ground in China. While the index for respondents in Denmark rose to 54.3 by the end of the year, confidence among managers in China fell to 48.51, dropping below the neutral level. Although the 15-day visa-free policy introduced in May made business travel easier and improved communication (Embassy of the People's Republic of China in Denmark, 2024), a significant difference in market perception remained. Furthermore, SMEs faced much harder conditions than large corporations. While big companies used their financial strength to keep up their R&D despite the price wars, many SMEs struggled with rising costs and intense local competition.

C.5 Year 2025

The DBOC index for 2025 shows a year with large swings in confidence, followed by a more stable ending. Danish companies' views on China changed noticeably over the year, moving from early optimism, to strong concern mid-year, and finally to a recovery in index. Overall, the curve tells a story of high expectations, disappointment, adjustment, and finally a more realistic outlook.

At the beginning of the year, Q1 2025 started on a positive note. The DBOC index rose to 52.92, which was the highest level since early 2023. This shows that many Danish companies expected better business conditions in China. Confidence improved among both companies based in China and those based in Denmark, and small and medium-sized companies (SMEs) were especially optimistic. Importantly, this positive view was formed before new US tariffs on China were fully announced and implemented, meaning expectations were still based on hope rather than real effects.

This optimism did not last long. In Q2 2025, the index dropped sharply to 48.96, falling below the neutral level of 50. This was the lowest point of the year and marked a clear change in mood. The fall happened across most company groups and sectors, showing that the problems were widespread. A key reason for this drop was growing concern about decoupling and weaker communication between China operations and company headquarters. Many firms reported missed projects and delayed decisions, and uncertainty began to affect daily operations. Q2 therefore reflects a moment where expectations were adjusted downward as challenges became more visible.

After this setback, the second half of the year shows a slow but steady improvement. In Q3 2025, the index increased to 50.78, moving back into slightly positive territory. This suggests that companies were beginning to adjust to the new situation rather than expecting conditions to quickly improve. The recovery was careful and uneven. Companies based in China showed more confidence than those based in Denmark, and SMEs rebounded strongly, while larger firms remained more cautious. Q3 can be seen as a phase of learning and adaptation rather than renewed optimism.

This stabilising trend continued in Q4 2025, when the index rose further to 51.40. Confidence strengthened again, especially among larger companies and firms based in Denmark. However, the level of the index remained moderate, showing that businesses were still careful. Towards the end of the year, companies increasingly focused on longer-term issues such as market access, sustainability, and supply-chain diversification, particularly in relation to China's upcoming 15th Five-Year Plan. This helped bring more clarity and slightly improve expectations.

Looking across the whole year, different groups experienced the changes differently. Companies operating in China were generally more stable in their outlook than those making decisions from Denmark. SMEs showed much stronger ups and downs, reacting quickly to both positive and negative signals, while larger firms moved more slowly but ended the year in a stronger position. At the sector level, sentiment also changed over time, but most sectors followed the same general path as the main

index: improvement early in the year, decline in mid-year, and mixed but more stable conditions toward the end.

C.6 Summary

This section sets the analytical context for the study by summarising the key developments observed in the Danish Business Outlook on China (DBOC) dataset from 2021 to 2025. Rather than providing a detailed year-by-year repetition of the data, the purpose of this section is to synthesise the main patterns emerging across the five-year period and to introduce the core analytical insights that guide the subsequent analysis. In doing so, the section establishes why organisational factors such as decoupling and broader geopolitical dynamics are central to understanding Danish companies' evolving perspectives on China.

Viewed as a whole, the five-year DBOC curve illustrates a clear transition in Danish companies' perceptions of China. The period begins in 2021 with strong optimism, reflecting China's continued importance as a market for sales, production, and growth opportunities. This optimism, however, gradually eroded as companies encountered increasing operational challenges and uncertainty, culminating in a sharp downturn during 2022.

From 2023 onward, the dataset reveals an uneven and cautious recovery. While moments of renewed optimism emerged most notably in early 2023 and early 2025, these were repeatedly followed by renewed uncertainty and adjustment. By 2024 and the last part of 2025, business confidence stabilised closer to neutral levels, suggesting a shift away from crisis-driven reactions toward a more realistic and risk-aware engagement with the Chinese market. Taken together, the curve reflects a movement from expectation-driven optimism to a phase characterised by learning, recalibration, and strategic caution.

One of the key insights emerging from this development is that external shocks alone cannot fully explain the observed patterns in the dataset. While global events clearly triggered downturns in confidence, the persistent volatility and divergence across companies point to the growing importance of internal organisational factors. Over time, how firms perceive and respond to external challenges increasingly depends on internal structures, coordination mechanisms, and decision-making processes. It is within this context that organisational decoupling becomes analytically important.

In this study, decoupling refers to the growing separation between strategic decisionmaking at headquarters and the operational realities faced by subsidiaries and managers on the ground in China. The DBOC data highlights this phenomenon through consistent differences in outlook between respondents based in Denmark and those based in China, as well as through increasing disparities between short-term expectations and longterm strategic commitments. These patterns suggest that uncertainty is not only externally imposed but also internally mediated. When communication

weakens or decisionmaking becomes slower and more fragmented, firms may experience more uncertainty even when underlying market conditions stabilise.

The role of geopolitical tensions further reinforces this dynamic. Unlike short-term operational disruptions, geopolitical tensions introduce structural uncertainty that reshapes firms' strategic horizons. Trade conflicts, regulatory scrutiny, and political signalling dose not only influence market access but also internal risk assessments and governance structures. As a result, companies must continuously balance global compliance requirements with local operational flexibility, intensifying internal coordination challenges. The dataset suggests that geopolitical factors increasingly shape expectations over the long term rather than causing isolated, short-lived disruptions.

Within this broader context, the life science sector represents a particularly relevant area of focus. life science firms operate at the intersection of commercial activity, public policy, and regulation, making them especially sensitive to geopolitical developments. At the same time, life science demand is relatively stable and long-term in nature, limiting firms' ability to withdraw quickly from key markets such as China. This we also see clear evidence of in the data set set forth by DBOC. This combination of strategic importance and regulatory exposure makes the life science sector an instructive case for analysing how companies adapt under sustained uncertainty.

Decoupling is especially consequential in the life science sector, where issues of safety, compliance, and regulatory approval require close coordination between headquarters and local operations. Increased organisational separation may elevate risks related to delayed decision-making, misaligned standards, or weakened oversight, while excessive centralisation may reduce local responsiveness. The sector therefore highlights how decoupling is not merely an abstract organisational challenge but a practical concern with direct implications for resilience and operational continuity.

Taken together, the patterns observed in the DBOC dataset provide a strong empirical foundation for the research focus of this study. The five-year evolution of business confidence illustrates not only how Danish companies' perspectives on China have changed, but also why questions of adaptation and business model resilience have become increasingly relevant. This makes the research question logically grounded in the dataset, as it seeks to explain both the observed shifts in perception and the strategic responses required to navigate a more complex and uncertain business environment.

Appendix D: AI Declaration

This project was developed with limited support from Generative Artificial Intelligence (GenAI), specifically OpenAI's ChatGPT (GPT-5.5), Anthropic's Claude and Moonshot AI's Kimi .

GenAI was used to assist with:

- Language refinement, including grammar correction, sentence structure, and improvement of academic writing style.
- Improving clarity and coherence of paragraphs and transitions between sections.
- Brainstorming and refining the structure of the literature review, theoretical framework, analysis, and discussion chapters.
- Assisting in the development and visual refinement of conceptual models and figures used in the project.
- Supporting the formulation of interview questions and the organization of qualitative findings.
- Supporting in the transcription process of the interviews.
- Providing feedback on argumentation, consistency, and readability throughout the writing process.

GenAI was not used as a source of empirical data, research findings, interview analysis, or final conclusions. All literature selection, source evaluation, data collection, coding, analysis, interpretation, and conclusions were conducted by the authors. All references were identified, verified, and cited by the authors, who take full responsibility for the accuracy, integrity, and originality of the final submission.

Prompts and AI-generated outputs were critically evaluated, modified, and integrated by the authors before inclusion in the project.

AI Tool Used: ChatGPT (OpenAI, GPT-5.5), Anthropic's Claude and Moonshot AI's Kimi
Period of Use: Spring Semester 2026
Extent of Use: Language refinement, structural support, transcription, conceptual discussion, and visual/figure development.